

Georgia Growth Comparison with IAPP District Removal: A Review of 2018 and 2019 Test Scores and SGPs.

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1 Background

In 2018, the Georgia legislature passed Senate Bill 362 which created a state innovative assessment pilot. SB 362 authorized the State Board of Education (SBOE) to approve up to 10 Georgia districts or consortia of districts to develop and implement innovative assessment systems. In August/September 2018, the SBOE held two rounds of applications for participants in the state pilot. In accordance with SB 362, the Georgia Department of Education (GaDOE) sought flexibility under the federal Innovative Assessment Demonstration Authority (IADA) authorized by ESEA Section 1204 to support the state's implementation of the innovative assessment pilot.

In July 2019, the U.S. Department of Education approved two Innovative Assessment Pilot Program (IAPP) consortia to develop and implement their assessment systems under the IADA. The Georgia MAP Assessment Partnership is partnering with NWEA to develop and implement MAP Growth for Georgia. The Putnam Consortium is partnering with Navvy Education, LLC to develop and implement Navvy. Both assessment systems include a series of formative or interim assessments administered throughout the school year that will roll up into a single, summative score.

The purpose of the IADA is to provide a five-year period during which the approved IAPP consortia can develop, implement, and scale their innovative assessment systems. Each approved consortium must demonstrate comparability between their innovative assessment system and the state's general academic assessments, Georgia Milestones, before being approved by GaDOE to implement their assessment system in lieu of Georgia Milestones. After such approval, all participating districts will have to continue to administer Georgia Milestones to the extent necessary (i.e., a sample of students) throughout the IADA period to continue to meet annual comparability requirements. Additionally, Georgia Milestones will need to be administered to all students in any grade, content area, or course for which the innovative assessment system does not have a corresponding assessment.

Once an IAPP consortium has demonstrated comparability between their innovative assessment system and Georgia Milestones and been approved by GaDOE, the innovative assessment system's results will be used in the state's College and Career Ready Performance Index (CCRPI) calculations for accountability purposes. Both consortia's assessment systems will produce a summative score, growth score, and literacy (Lexile) score that will be used in these calculations.

During the five-year pilot period, the GaDOE will manage the demonstration authority, collect stakeholder feedback, and oversee an external technical evaluation of the innovative assessment systems. At the end of the demonstration authority (the close of the 2023-2024 school year), the state will determine whether either of the innovative assessment systems should be administered statewide. Georgia is required to determine which one assessment it will administer to all students by the end of its participation in the IADA.

2 Analysis Workflow

In order to assess the potential impact of removing IAPP consortia students on Student Growth Percentile (SGP) results used to make Georgia accountability decisions, we reanalyzed the 2018 and 2019 Georgia Milestones end-of-grade (EOG) and end-of-course (EOC) test

data excluding consortia member student records. The same course progressions, number of prior test scores used, and all other relevant SGP model configurations were applied to these comparison analyses.

This report serves as an exploratory that provides statistical summaries and visualizations to describe the changes in individual and school level SGP results due to changes in student norm groups. IAPP consortia represent roughly 10% of Georgia's student population. In the subsections below, details for each content area and grade level (for EOG subjects) are provided. The Ranked SIMEX measurement error corrected SGP estimates are used for all summaries and visualizations as this value is the official growth metric used by the Georgia Department of Education for accountability purposes. As with other reports provided on the SGP analyses, only schools and grade-levels with a minimum of 15 students are used in these summaries.

The report is broken down into content area specific sections. The EOG ELA and Mathematics sections begin with school-level aggregations of achievement and growth. This summary level view is important to consider as they are more relevant to measures Georgia uses to make accountability judgements. In these summaries, the Milestones scale scores have been standardized by grade in order to make appropriate cross-grade comparisons. Subsequently summaries for each grade-level are provided. This gives a more detailed and appropriate comparison for growth as SGP analyses are calculated for each unique year/content area/grade combination. The EOC sections are presented by content area only as student grade-level is ignored in those analyses.

2.1 IAPP vs Non-IAPP Comparability

Comparing the results of the historical and revised SGP results is benefited by first comparing the IAPP and Non-IAPP students and schools. To provide this context, the first page of each subsection below provides summaries of the student- and school-level current achievement and growth. This information allows us to investigate these questions:

- **Do IAPP students and schools have different levels of achievement than other Georgia students and schools?**
- **Do IAPP students and schools have different levels of growth than other Georgia students and schools?**

A set of four distribution plots are provided for each comparison in a 2 by 2 matrix. The top rows contain information about individual students and the bottom rows are schools. The left side columns look at achievement (Milestones scale scores) and growth is in the right column. School level data in these cases are the mean scores and mean ranked SIMEX SGPs. These plots allow us to see if there are clear differences in the distributions between the two groups and the degree to which these values are consistent from one year to the next within the groups. In the case of the individual student growth percentile distribution (upper-right plot), we expect to see a uniform (box-like) distribution for the non-IAPP students and look for deviations from that in the IAPP students (dashed lines). In the other three plots we expect more normal (bell-curve) distributions, paying attention to deviations between the groups, as well as between years within the groups.

Summary statics are provided in tables as well to give additional context such as the median values and the number of students and schools.

2.1.1 Consortia Group Demographic Composition

The demographic composition of the IAPP consortia and non-consortia groups is another important factor to consider in addition to their achievement and growth. Table 1 presents the percent of students categorized as 1) economically disadvantaged, 2) having a disability and 3) English language learners. As the table shows, the two groups have similar numbers of students with disabilities and English learners, but the IAPP consortia have approximately 15% more economically disadvantaged students in their population.

Table 1: Demographic Composition of Consortia Groups by Year

Group	Year	Econ Disadv		Disabilities		Eng Learners		N
		No	Yes	No	Yes	No	Yes	
IAPP	2018	22.4	77.6	95.5	4.5	88.4	11.6	178,493
	2019	21.7	78.3	94.8	5.2	88.3	11.7	178,655
Non-IAPP	2018	37.3	62.7	95.8	4.2	87.8	12.2	1,539,501
	2019	37.9	62.1	95.0	5.0	87.4	12.6	1,542,266

2.2 Comparability of Growth Results with IAPP Students Removed

Given any differences observed between IAPP and non-IAPP students and schools, we would expect to also see some differences in the student growth estimates if a significant number of non-random students are removed from the norm groups from which they are calculated. The second page of each subsection includes summary tables and visualizations of these differences at the student- and school-levels.

These summaries help to address questions such as:

- If IAPP students and schools are different, what differences in growth might we see going forward?
- What magnitude of changes in growth might we see going forward if IAPP consortia students are omitted from SGP analyses?
- What type of school might be most impacted by removing IAPP consortia members?

Here we first provide a summary table of the differences in the SGP estimates at the student-level, where the difference is defined as the comparison SGP estimate from the reduced cohort minus the original SGP. School-level differences are similarly defined as the original mean SGP (MSGP) subtracted from the revised comparison MSGP. Visualizations provided in this section are for the distributions of original and revised school-level MSGPs for 2018 and 2019, and a scatterplot of the relationship between MSGP difference by school size. A summary table for the MSGP differences is provided for additional context.

3 School Level ELA Results

Figure 1: 2018 and 2019 School Level ELA Consortium Member Comparisons

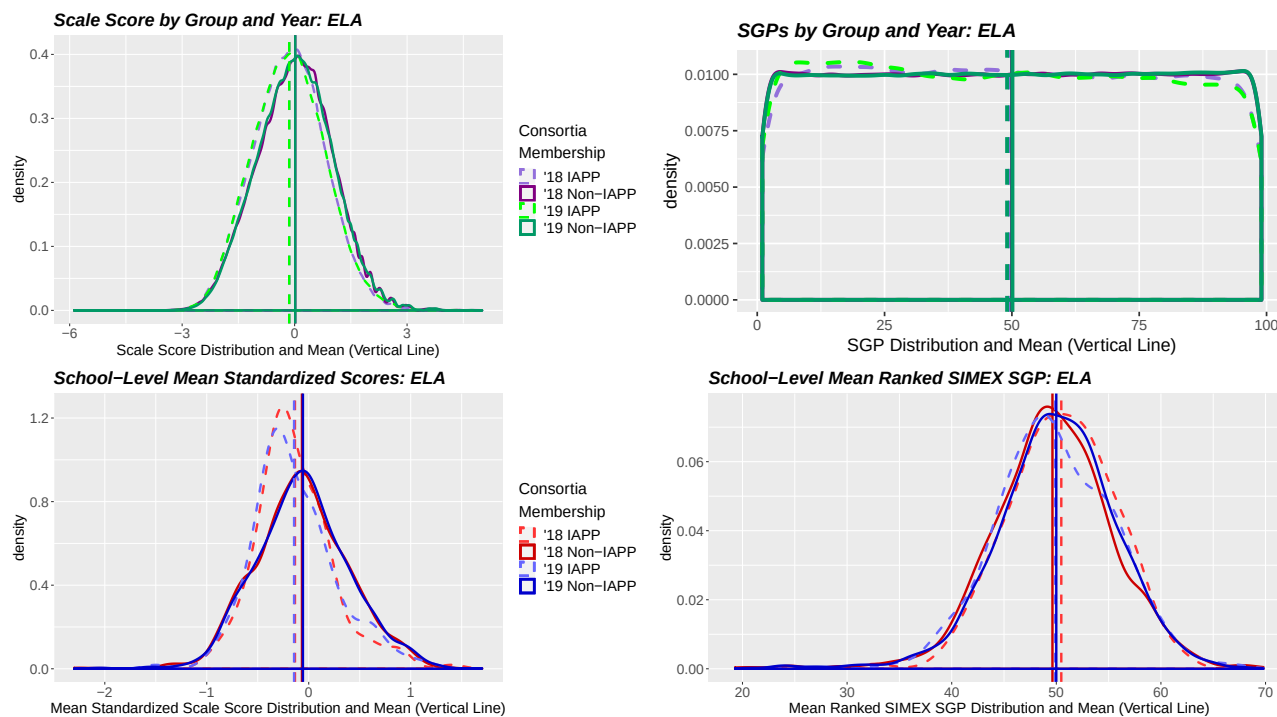


Table 2: Mean and Median Values for School Level ELA - Complete Georgia Analysis.

Group	Year	Z Scores		SGPs		N
		Mean	Median	Mean	Median	
IAPP Students	2018	-0.14	-0.11	50.1	50	64,715
	2019	-0.14	-0.15	49.6	49	65,269
Non-IAPP Students	2018	0.02	0.01	50.1	50	554,959
	2019	0.02	0.02	50.1	50	559,413
IAPP School Means	2018	-0.13	-0.18	50.5	51	177
	2019	-0.14	-0.22	49.9	50	179
Non-IAPP School Means	2018	-0.06	-0.09	49.6	50	1,579
	2019	-0.06	-0.07	50.0	50	1,584

3.1 ELA IAPP Consortium Exclusion Comparison

Table 3: Differences in Student Level ELA SGPs (Non-IAPP Results minus Complete).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-34	-1	0	-0.01	1	52
2019	-50	-1	0	-0.09	1	58

Figure 2: 2018 and 2019 ELA Alternate Analysis Comparison - School Level Mean SGPs (MSGP)

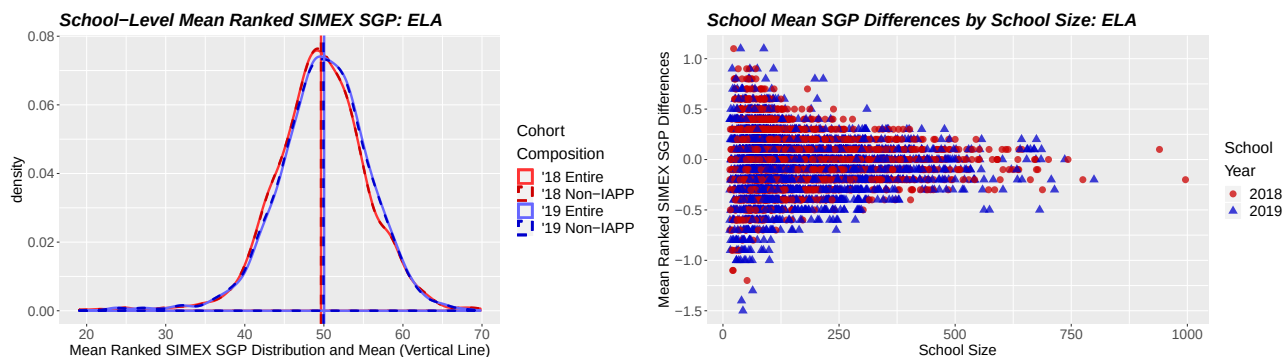


Table 4: Differences in School Level ELA Mean SGPs (Non-IAPP District MSGP minus Complete MSGP).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-0.8	-0.1	0.0	0.01	0.1	0.6
2019	-1.0	-0.2	-0.1	-0.09	0.0	1.0

The IAPP students and schools show overall lower achievement in ELA and their growth is very similar. Although omitting IAPP students from the SGP norm group has a significant impact on some individual students' growth estimates, there is minimal change in the school-level MSGPs. Differences in these aggregates range from -1 to 1, which is small when compared to the annual fluctuations in MSGPs for most schools. The impact of removing IAPP students also decreases substantially with increasing school size.

4 Grade Level ELA Results

4.1 4th Grade ELA by Consortium Membership

Figure 3: 2018 and 2019 4th Grade ELA Consortium Member Comparisons

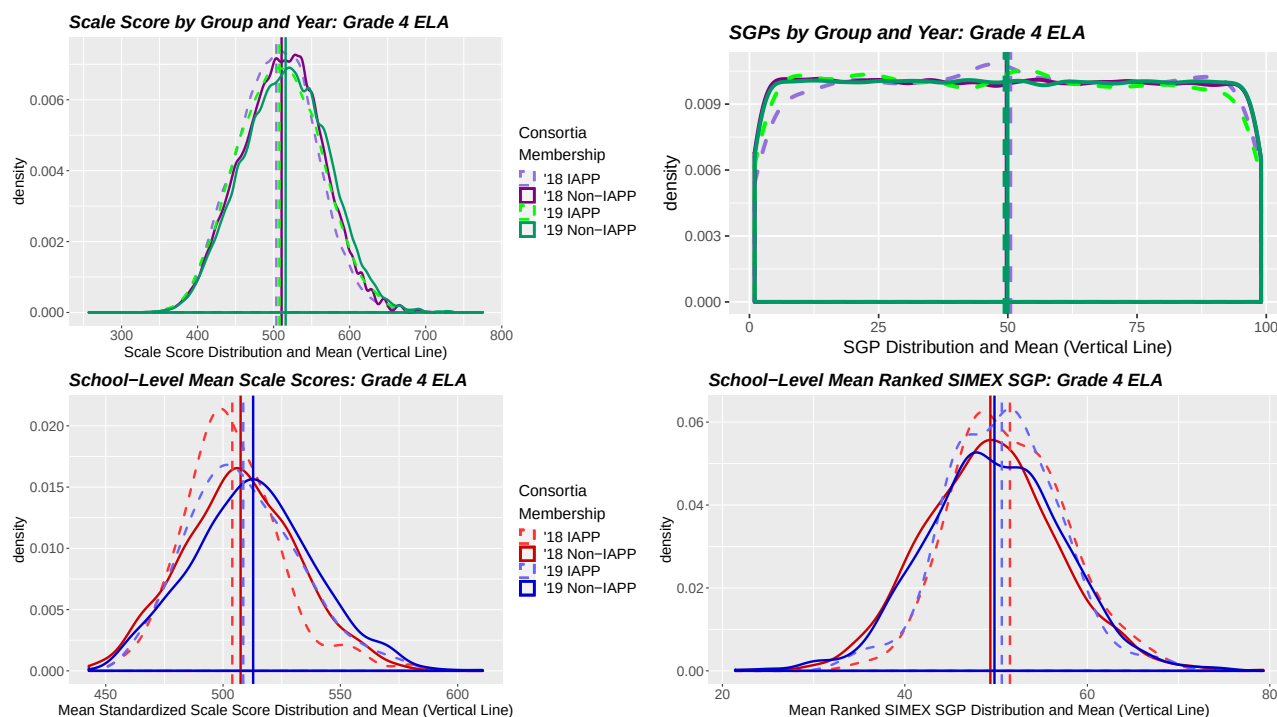


Table 5: Mean and Median Values for 4th Grade ELA - Complete Georgia Analysis.

Group	Year	Scale Scores		SGPs		N
		Mean	Median	Mean	Median	
IAPP Students	2018	503.6	504.0	51.2	51	13,507
	2019	507.7	507.0	50.0	50	12,963
Non-IAPP Students	2018	510.4	510.0	49.9	50	114,081
	2019	515.8	517.0	50.0	50	112,511
IAPP School Means	2018	503.9	502.5	51.5	51	123
	2019	508.6	507.0	50.6	51	124
Non-IAPP School Means	2018	507.5	509.0	49.4	49	1,113
	2019	512.8	513.0	49.8	50	1,116

4.2 4th Grade ELA IAPP Consortium Exclusion Comparison

Table 6: Differences in Student Level 4th Grade ELA SGPs (Non-IAPP Results minus Complete).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-14	-1	0	0.11	1	21
2019	-14	-1	0	-0.04	1	10

Figure 4: 2018 and 2019 4th Grade ELA Alternate Analysis Comparison - School Level Mean SGPs (MSGP)

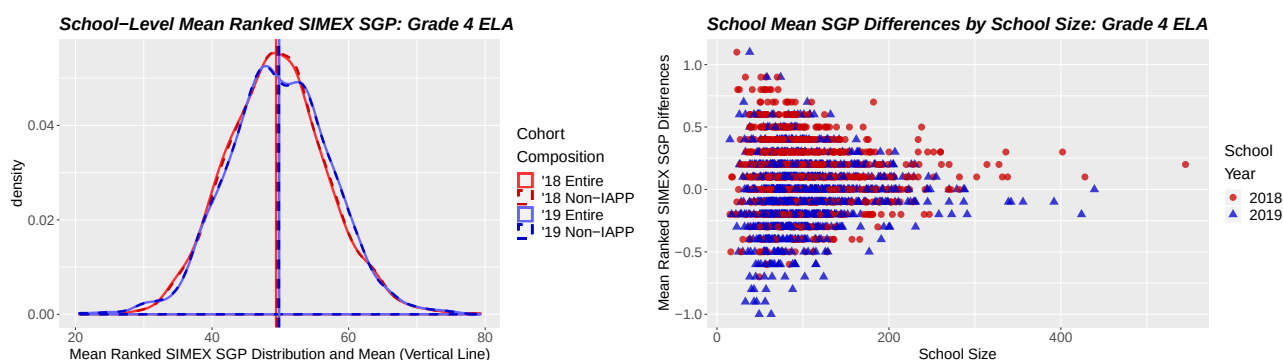


Table 7: Differences in School Level 4th Grade ELA Mean SGPs (Non-IAPP District MSGP minus Complete MSGP).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-0.7	-0.1	0.1	0.12	0.3	1.1
2019	-1.0	-0.2	0.0	-0.05	0.1	1.1

The IAPP students and schools show overall lower achievement in Grade 4 ELA and have slightly higher growth. Although omitting IAPP students from the SGP norm group impacts some individual students' growth estimates, there is minimal change in the school-level MSGPs. Differences in these aggregates range from -1 to 1, which is small when compared to the annual fluctuations in MSGPs for most schools. The impact of removing IAPP students also decreases substantially with increasing school size.

It should be noted that Grade 4 ELA analyses use only a single prior score (Grade 3 ELA) unlike most end-of-grade analyses which use up to two priors. This likely decreases the impact of removing a substantial part of the statewide norm group.

4.3 5th Grade ELA by Consortium Membership

Figure 5: 2018 and 2019 5th Grade ELA Consortium Member Comparisons

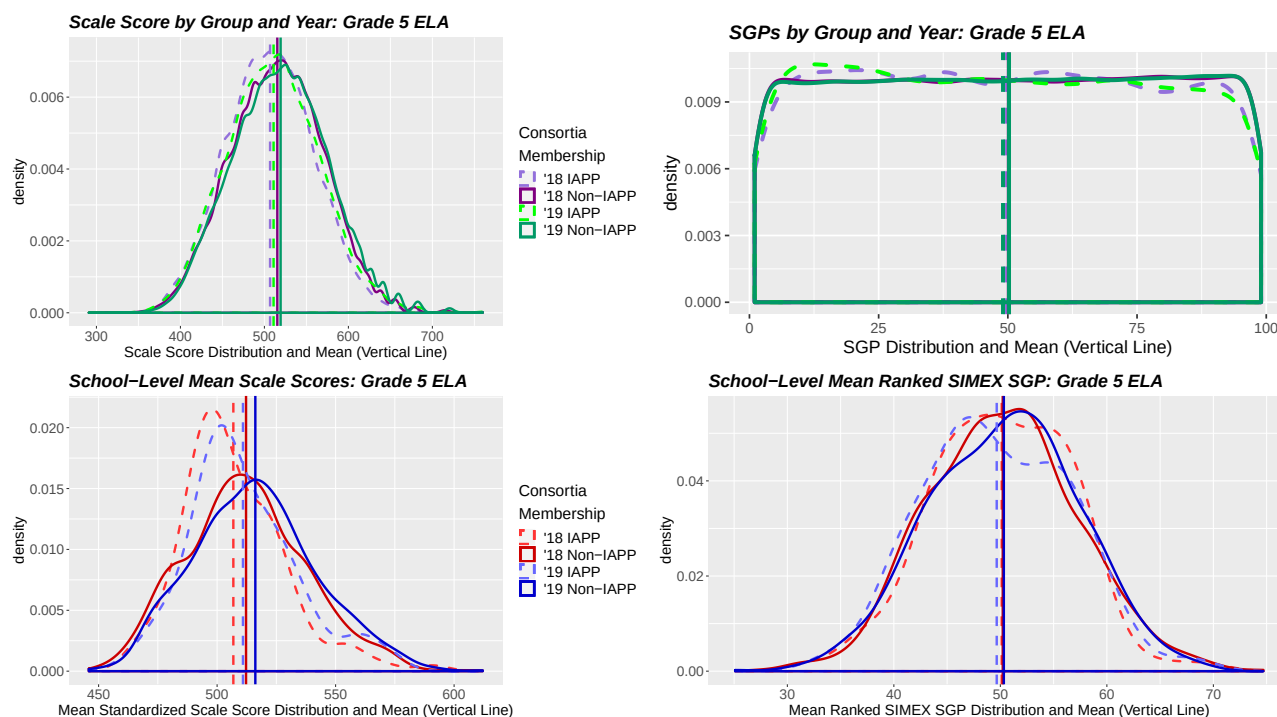


Table 8: Mean and Median Values for 5th Grade ELA - Complete Georgia Analysis.

Group	Year	Scale Scores		SGPs		N
		Mean	Median	Mean	Median	
IAPP Students	2018	506.7	504.0	49.9	50.0	13,519
	2019	510.9	511.0	49.4	49.0	13,636
Non-IAPP Students	2018	515.4	514.0	50.2	50.0	115,168
	2019	519.3	518.0	50.3	50.0	115,219
IAPP School Means	2018	506.9	504.0	50.1	50.5	123
	2019	510.9	506.0	49.7	49.0	124
Non-IAPP School Means	2018	512.2	511.5	50.2	50.0	1,103
	2019	516.1	516.0	50.3	50.5	1,108

4.4 5th Grade ELA IAPP Consortium Exclusion Comparison

Table 9: Differences in Student Level 5th Grade ELA SGPs (Non-IAPP Results minus Complete).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-28	-1	0	-0.05	1	22
2019	-44	-1	0	-0.12	1	45

Figure 6: 2018 and 2019 5th Grade ELA Alternate Analysis Comparison - School Level Mean SGPs (MSGP)

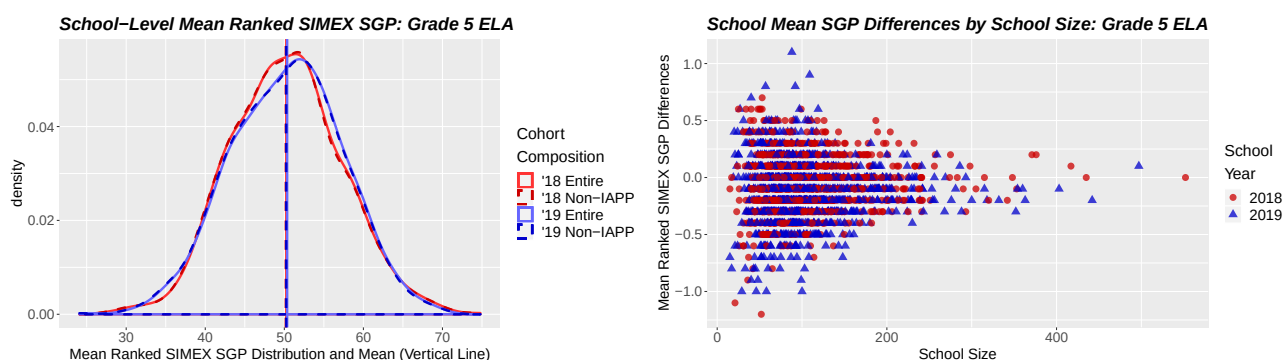


Table 10: Differences in School Level 5th Grade ELA Mean SGPs (Non-IAPP District MSGP minus Complete MSGP).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-1.2	-0.2	-0.1	-0.06	0.1	0.7
2019	-1.0	-0.3	-0.1	-0.12	0.0	1.1

The IAPP students and schools show overall lower achievement in Grade 5 ELA and slightly lower growth. Although omitting IAPP students from the SGP norm group impacts some individual students' growth estimates substantially (more so than Grade 4 ELA), there is minimal change in the school-level MSGPs. Differences in these aggregates range from approximately -1 to 1, which is small when compared to the annual fluctuation in MSGPs for most schools. The impact of removing IAPP students also decreases substantially with increasing school size.

4.5 6th Grade ELA by Consortium Membership

Figure 7: 2018 and 2019 6th Grade ELA Consortium Member Comparisons

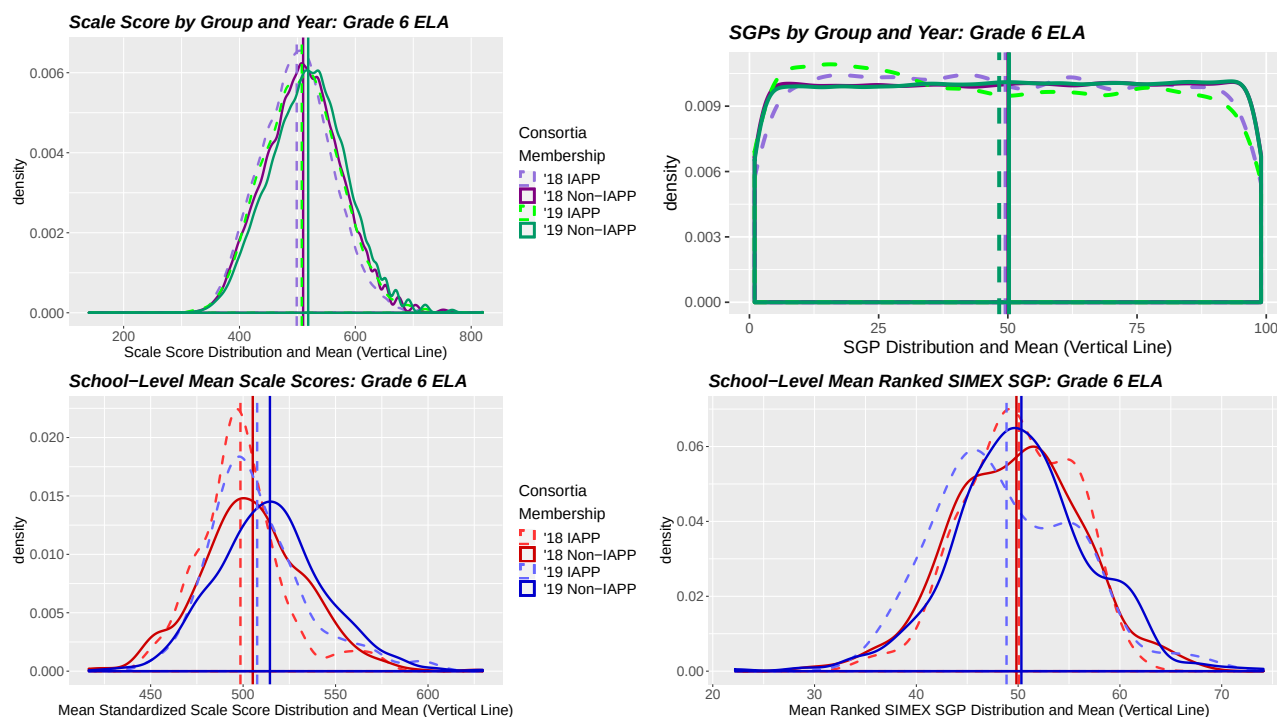


Table 11: Mean and Median Values for 6th Grade ELA - Complete Georgia Analysis.

Group	Year	Scale Scores		SGPs		N
		Mean	Median	Mean	Median	
IAPP Students	2018	498.8	501.0	50.0	50	13,075
	2019	507.6	508.0	48.7	48	13,425
Non-IAPP Students	2018	509.9	512.0	50.1	50	111,296
	2019	518.6	520.0	50.3	50	114,822
IAPP School Means	2018	498.5	496.5	50.1	49	52
	2019	507.6	503.0	48.9	47	52
Non-IAPP School Means	2018	505.2	503.0	49.8	50	515
	2019	514.5	515.0	50.3	50	513

4.6 6th Grade ELA IAPP Consortium Exclusion Comparison

Table 12: Differences in Student Level 6th Grade ELA SGPs (Non-IAPP Results minus Complete).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-34	-1	0	-0.02	1	52
2019	-37	-1	0	-0.19	1	39

Figure 8: 2018 and 2019 6th Grade ELA Alternate Analysis Comparison - School Level Mean SGPs (MSGP)

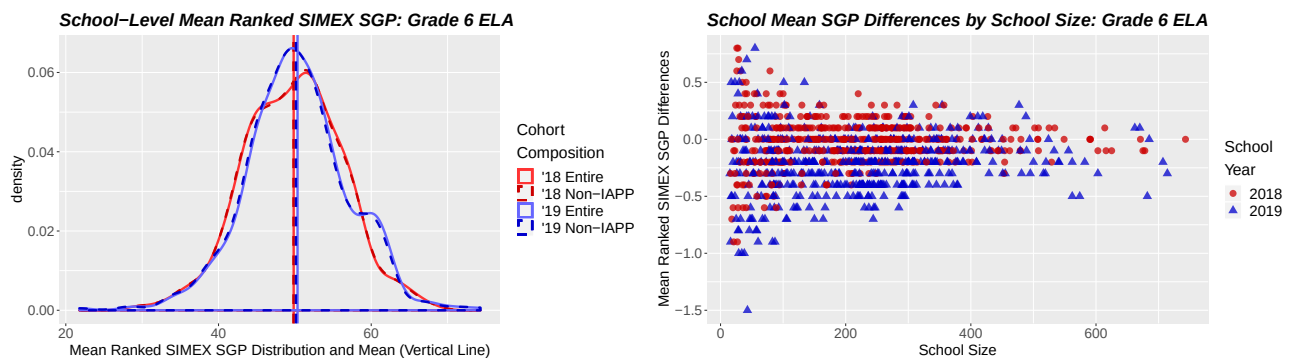


Table 13: Differences in School Level 6th Grade ELA Mean SGPs (Non-IAPP District MSGP minus Complete MSGP).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-0.9	-0.1	0.0	-0.02	0.1	0.8
2019	-1.5	-0.4	-0.2	-0.21	-0.1	0.8

The IAPP students and schools show overall lower achievement in Grade 6 ELA. The typical growth is generally similar, although IAPP students and schools both saw a decrease in growth in 2019. Although omitting IAPP students from the SGP norm group impacts some individual students' growth estimates substantially, there is minimal change in the school-level MSGPs. Differences in these aggregates range from -1.5 to 1, which is small when compared to the annual fluctuation in MSGPs for most schools. As with the previous grades reviewed above, the impact of removing IAPP students decreases substantially with increasing school size.

4.7 7th Grade ELA by Consortium Membership

Figure 9: 2018 and 2019 7th Grade ELA Consortium Member Comparisons

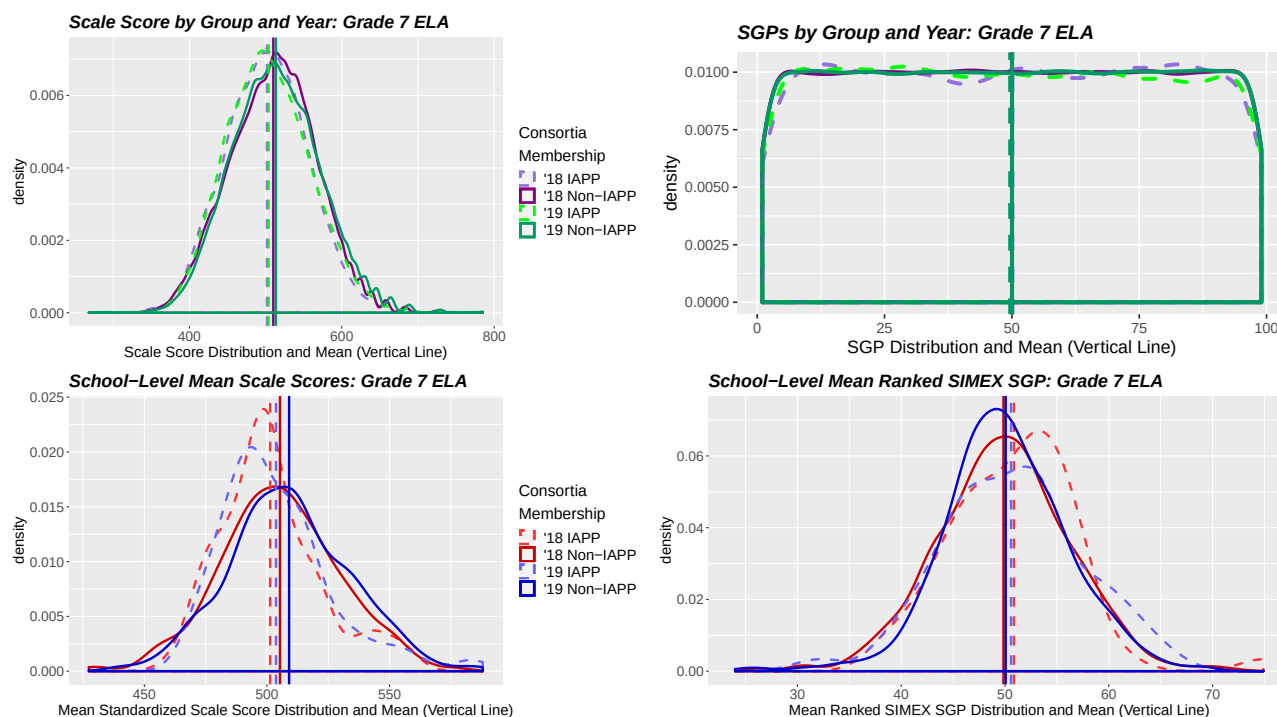


Table 14: Mean and Median Values for 7th Grade ELA - Complete Georgia Analysis.

Group	Year	Scale Scores		SGPs		N
		Mean	Median	Mean	Median	
IAPP Students	2018	501.7	503	50.5	51.0	12,634
	2019	503.8	503	50.2	50.0	13,075
Non-IAPP Students	2018	510.1	513	50.0	50.0	109,324
	2019	513.4	513	50.1	50.0	112,047
IAPP School Means	2018	501.3	498	50.9	52.2	52
	2019	503.7	498	50.6	51.0	52
Non-IAPP School Means	2018	505.3	504	49.9	50.0	494
	2019	509.0	508	50.1	50.0	501

4.8 7th Grade ELA IAPP Consortium Exclusion Comparison

Table 15: Differences in Student Level 7th Grade ELA SGPs (Non-IAPP Results minus Complete).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-18	-1	0	0.04	1	39
2019	-41	-1	0	0.01	1	45

Figure 10: 2018 and 2019 7th Grade ELA Alternate Analysis Comparison - School Level Mean SGPs (MSGP)

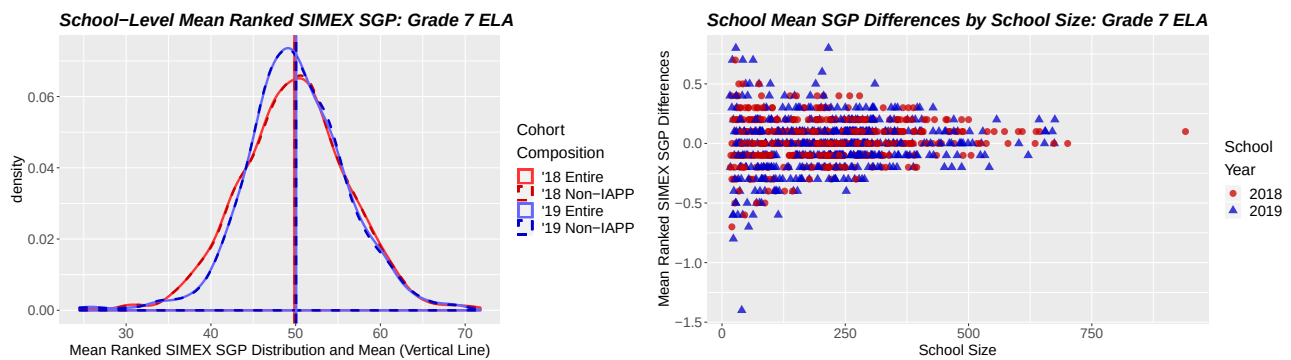


Table 16: Differences in School Level 7th Grade ELA Mean SGPs (Non-IAPP District MSGP minus Complete MSGP).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-0.7	-0.1	0	0.03	0.1	0.7
2019	-1.4	-0.1	0	-0.01	0.1	0.8

The IAPP students and schools show slightly lower achievement in Grade 7 ELA. Growth is similar between the two groups, although IAPP students and schools slightly outperform their non-IAPP peers.

Although omitting IAPP students from the SGP norm group impacts some individual students' growth estimates substantially, there is minimal change in the school-level MSGPs. Differences in these aggregates range from approximately -1.5 to 1, which is small when compared to the annual fluctuation in MSGPs for most schools. As with the other ELA grades, the impact of removing IAPP students decreases substantially with increasing school size.

4.9 8th Grade ELA by Consortium Membership

Figure 11: 2018 and 2019 8th Grade ELA Consortium Member Comparisons

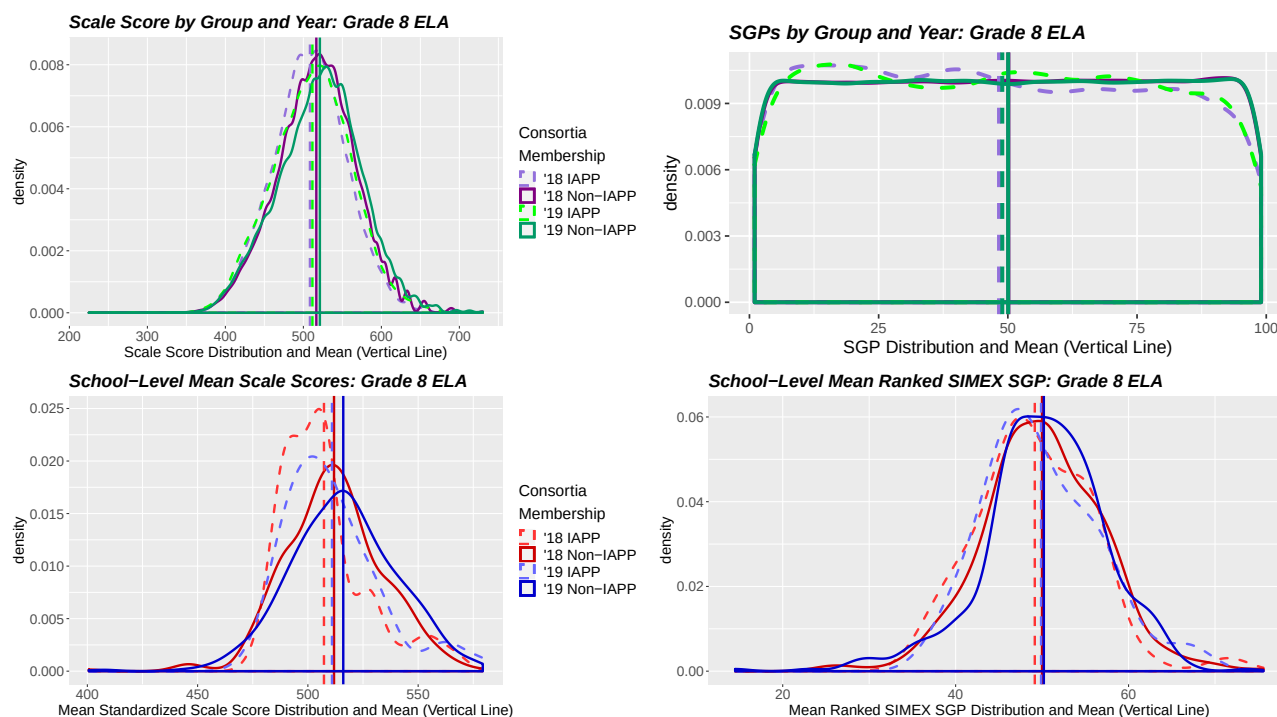


Table 17: Mean and Median Values for 8th Grade ELA - Complete Georgia Analysis.

Group	Year	Scale Scores		SGPs		N
		Mean	Median	Mean	Median	
IAPP Students	2018	508.4	507	48.7	48.0	11,980
	2019	511.6	514	49.4	50.0	12,170
Non-IAPP Students	2018	516.6	516	50.1	50.0	105,090
	2019	521.2	524	50.1	50.0	104,814
IAPP School Means	2018	507.3	507	49.2	49.2	52
	2019	510.9	509	49.9	49.5	52
Non-IAPP School Means	2018	511.9	511	50.0	50.0	491
	2019	516.0	516	50.2	51.0	493

4.10 8th Grade ELA IAPP Consortium Exclusion Comparison

Table 18: Differences in Student Level 8th Grade ELA SGPs (Non-IAPP Results minus Complete).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-30	-1	0	-0.16	1	14
2019	-50	-1	0	-0.10	1	58

Figure 12: 2018 and 2019 8th Grade ELA Alternate Analysis Comparison - School Level Mean SGPs (MSGP)

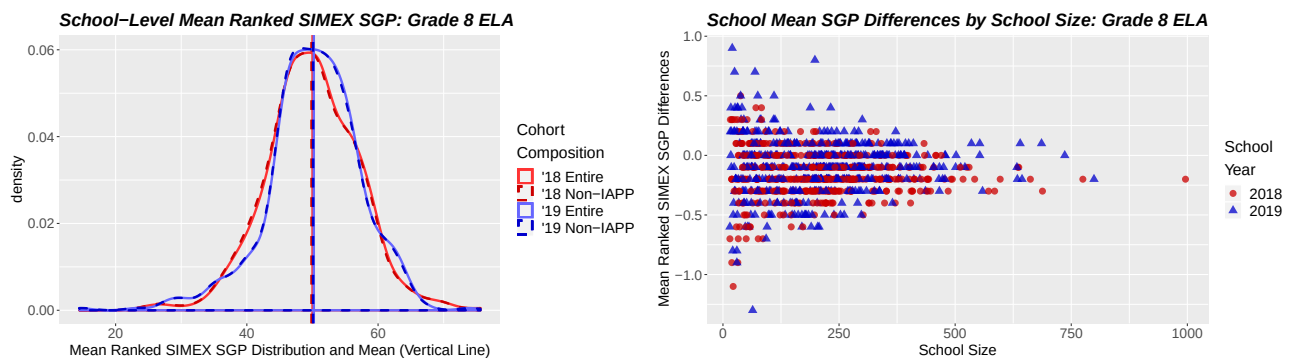


Table 19: Differences in School Level 8th Grade ELA Mean SGPs (Non-IAPP District MSGP minus Complete MSGP).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-1.1	-0.3	-0.2	-0.16	0	0.5
2019	-1.3	-0.2	-0.1	-0.10	0	0.9

The IAPP students and schools show lower achievement in Grade 8 ELA. The typical growth is similar, with IAPP students and schools slightly underperforming compared to their non-IAPP peers.

Although omitting IAPP students from the SGP norm group impacts some individual students' growth estimates substantially, there is minimal change in the school-level MSGPs. Differences in these aggregates range from approximately -1.5 to 1, which is small when compared to the annual fluctuation in MSGPs for most schools. The impact of removing IAPP students also decreases substantially with increasing school size.

5 School Level Mathematics Results

Figure 13: 2018 and 2019 School Level Math Consortium Member Comparisons

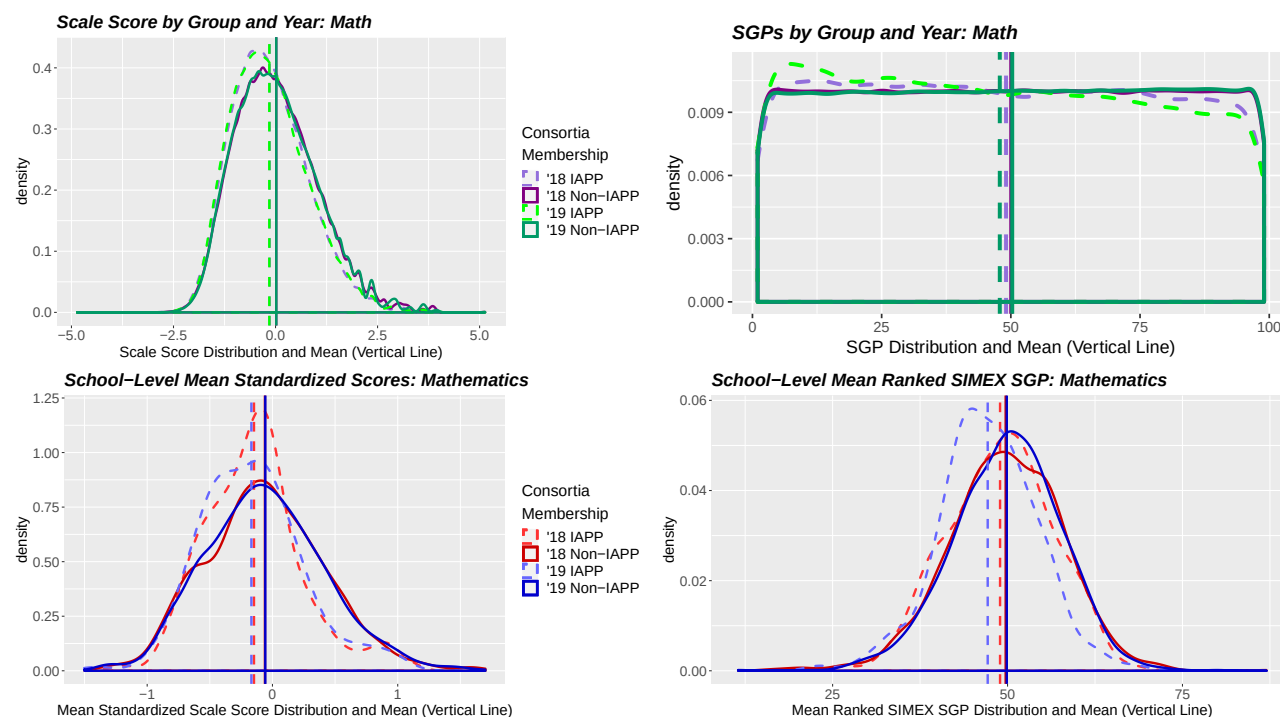


Table 20: Mean and Median Values for School Level Math - Complete Georgia Analysis.

Group	Year	Z Scores		SGPs		N
		Mean	Median	Mean	Median	
IAPP Students	2018	-0.14	-0.24	49.4	49	63,699
	2019	-0.15	-0.23	48.2	47	63,987
Non-IAPP Students	2018	0.02	-0.08	50.1	50	537,776
	2019	0.02	-0.07	50.3	50	539,229
IAPP School Means	2018	-0.15	-0.25	48.9	49	177
	2019	-0.16	-0.23	47.2	46	179
Non-IAPP School Means	2018	-0.06	-0.14	49.7	50	1,580
	2019	-0.06	-0.15	49.9	50	1,583

5.1 Mathematics IAPP Consortium Exclusion Comparison

Table 21: Differences in Student Level Math SGPs (Non-IAPP Results minus Complete).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-33	-1	0	-0.11	1	18
2019	-51	-1	0	-0.25	1	49

Figure 14: 2018 and 2019 Math Alternate Analysis Comparison - School Level Mean SGPs (MSGP)

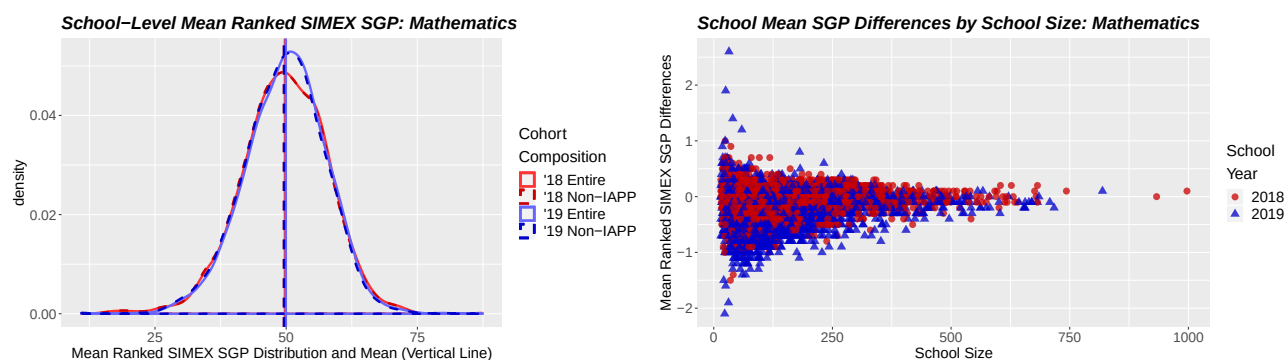


Table 22: Differences in School Level Math Mean SGPs (Non-IAPP District MSGP minus Complete MSGP).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-0.9	-0.3	-0.2	-0.18	0.0	0.7
2019	-2.1	-0.5	-0.4	-0.35	-0.2	0.7

The IAPP students and schools show overall lower achievement in Math and their growth is lower, particularly in 2019. Although omitting IAPP students from the SGP norm group has a significant impact on some individual students' growth estimates, there is minimal change in the school-level MSGPs. Differences in these aggregates range from approximately -2 to 1, which is small when compared to the annual fluctuation in MSGPs for most schools. The impact of removing IAPP students also decreases substantially with increasing school size.

6 Grade Level Math Results

6.1 4th Grade Mathematics by Consortium Membership

Figure 15: 2018 and 2019 4th Grade Mathematics Consortium Member Comparisons

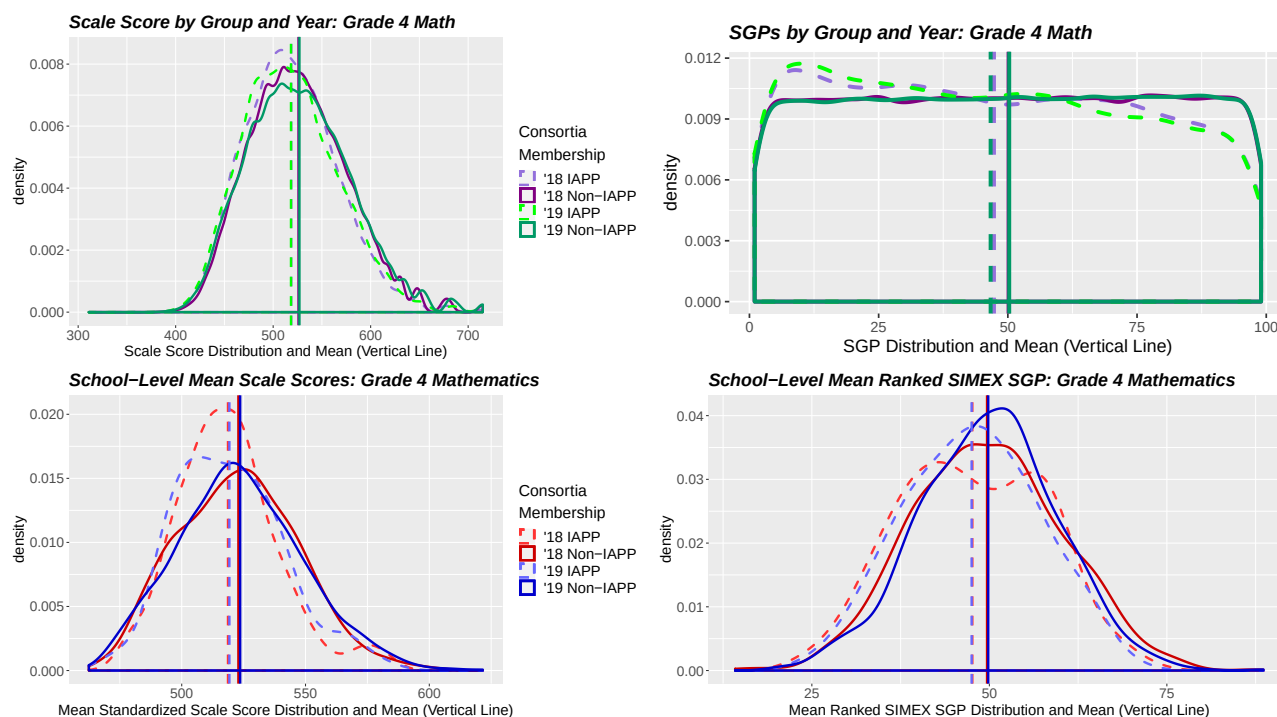


Table 23: Mean and Median Values for 4th Grade Math - Complete Georgia Analysis.

Group	Year	Scale Scores		SGPs		N
		Mean	Median	Mean	Median	
IAPP Students	2018	518.6	517	47.8	47.0	13,525
	2019	518.3	514	47.1	46.0	12,954
Non-IAPP Students	2018	526.0	523	50.3	50.0	114,464
	2019	527.0	525	50.3	51.0	112,410
IAPP School Means	2018	518.7	513	47.6	46.0	123
	2019	519.2	514	47.5	46.2	124
Non-IAPP School Means	2018	522.9	520	49.7	49.0	1,113
	2019	523.6	521	49.8	50.0	1,116

6.2 4th Grade Math IAPP Consortium Exclusion Comparison

Table 24: Differences in Student Level 4th Grade Math SGPs (Non-IAPP Results minus Complete).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-10	-1	0	-0.34	1	7
2019	-11	-1	0	-0.34	1	11

Figure 16: 2018 and 2019 4th Grade Math Alternate Analysis Comparison - School Level Mean SGPs (MSGP)

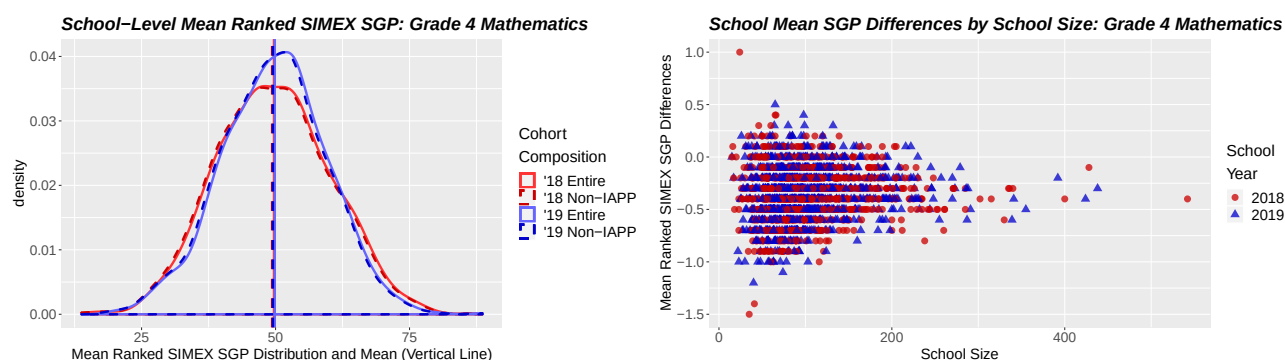


Table 25: Differences in School Level 4th Grade Math Mean SGPs (Non-IAPP District MSGP minus Complete MSGP).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-1.5	-0.5	-0.3	-0.35	-0.2	1.0
2019	-1.2	-0.5	-0.4	-0.36	-0.2	0.5

The IAPP students and schools show overall lower achievement in Grade 4 Math and they underperform in growth as well. Although omitting IAPP students from the SGP norm group impacts some individual students' growth estimates, there is minimal change in the school-level MSGPs. Differences in these aggregates range from approximately -1.5 to 1, which is much lower than annual fluctuations in MSGPs for most schools. The impact of removing IAPP students also decreases substantially with increasing school size.

It should be noted that Grade 4 ELA analyses use only a single prior score (Grade 3 ELA) unlike most end-of-grade analyses which use up to two priors. This likely decreases the impact of removing a substantial part of the statewide norm group.

6.3 5th Grade Mathematics by Consortium Membership

Figure 17: 2018 and 2019 5th Grade Mathematics Consortium Member Comparisons

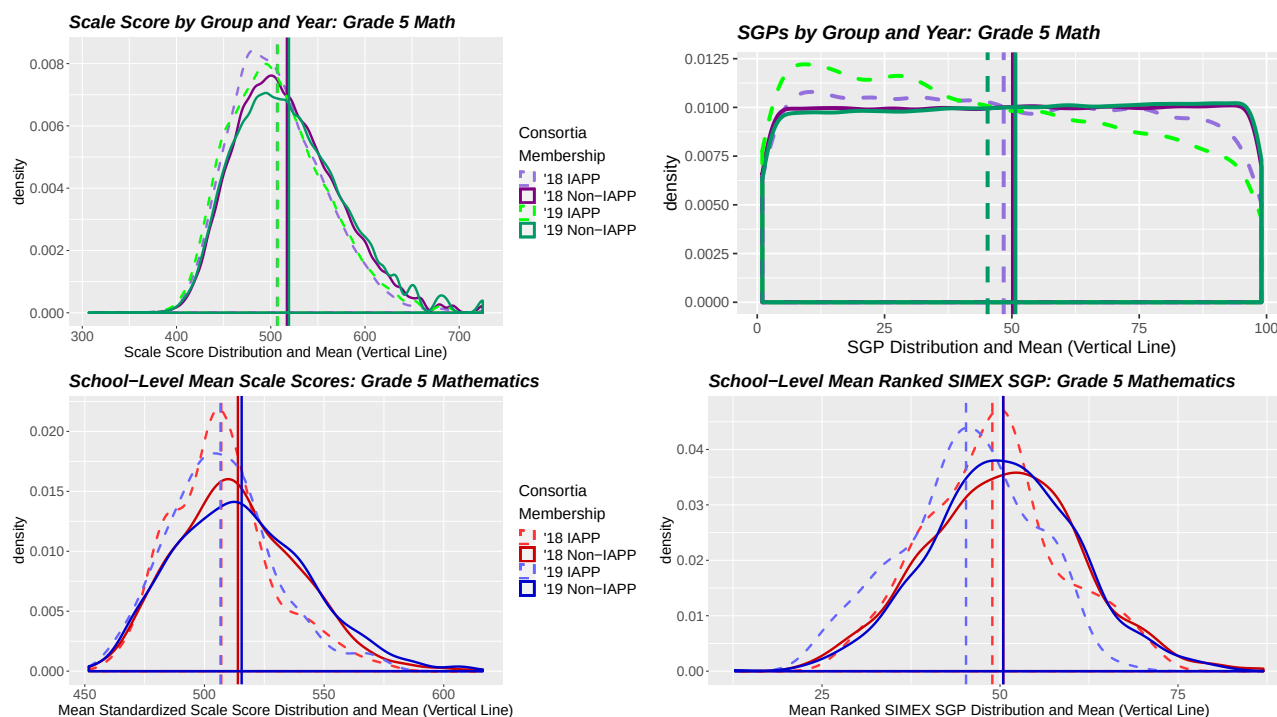


Table 26: Mean and Median Values for 5th Grade Math - Complete Georgia Analysis.

Group	Year	Scale Scores		SGPs		N
		Mean	Median	Mean	Median	
IAPP Students	2018	506.7	502	48.7	48.0	13,555
	2019	507.3	502	45.6	44.0	13,619
Non-IAPP Students	2018	517.1	512	50.2	50.0	115,575
	2019	519.4	512	50.7	51.0	115,138
IAPP School Means	2018	507.0	502	48.9	48.0	123
	2019	506.7	498	45.2	42.5	124
Non-IAPP School Means	2018	514.0	509	50.5	51.0	1,103
	2019	515.5	509	50.5	50.5	1,108

6.4 5th Grade Math IAPP Consortium Exclusion Comparison

Table 27: Differences in Student Level 5th Grade Math SGPs (Non-IAPP Results minus Complete).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-14	-1	0	-0.18	1	9
2019	-35	-2	0	-0.57	0	40

Figure 18: 2018 and 2019 5th Grade Math Alternate Analysis Comparison - School Level Mean SGPs (MSGP)

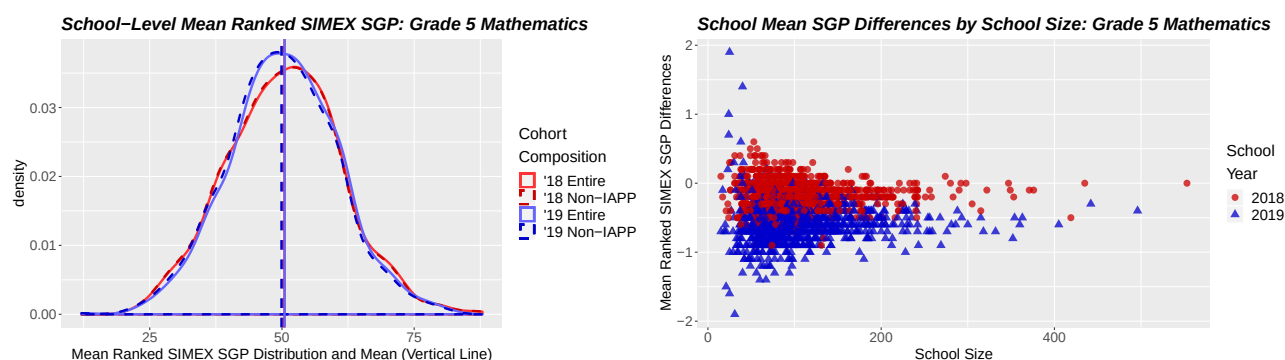


Table 28: Differences in School Level 5th Grade Math Mean SGPs (Non-IAPP District MSGP minus Complete MSGP).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-0.9	-0.3	-0.2	-0.17	0.0	0.6
2019	-1.9	-0.7	-0.6	-0.57	-0.4	1.9

The IAPP students and schools show overall lower achievement in Grade 5 Math and they underperform in growth as well (particularly in 2019). Although omitting IAPP students from the SGP norm group impacts some individual students' growth estimates substantially, there is minimal change in the school-level MSGPs. Differences in these aggregates range from approximately -2 to 2, which is much lower than annual fluctuations in MSGPs for most schools. The impact of removing IAPP students also decreases substantially with increasing school size. The change is particularly large in 2019 given low IAPP growth in that year.

6.5 6th Grade Mathematics by Consortium Membership

Figure 19: 2018 and 2019 6th Grade Mathematics Consortium Member Comparisons

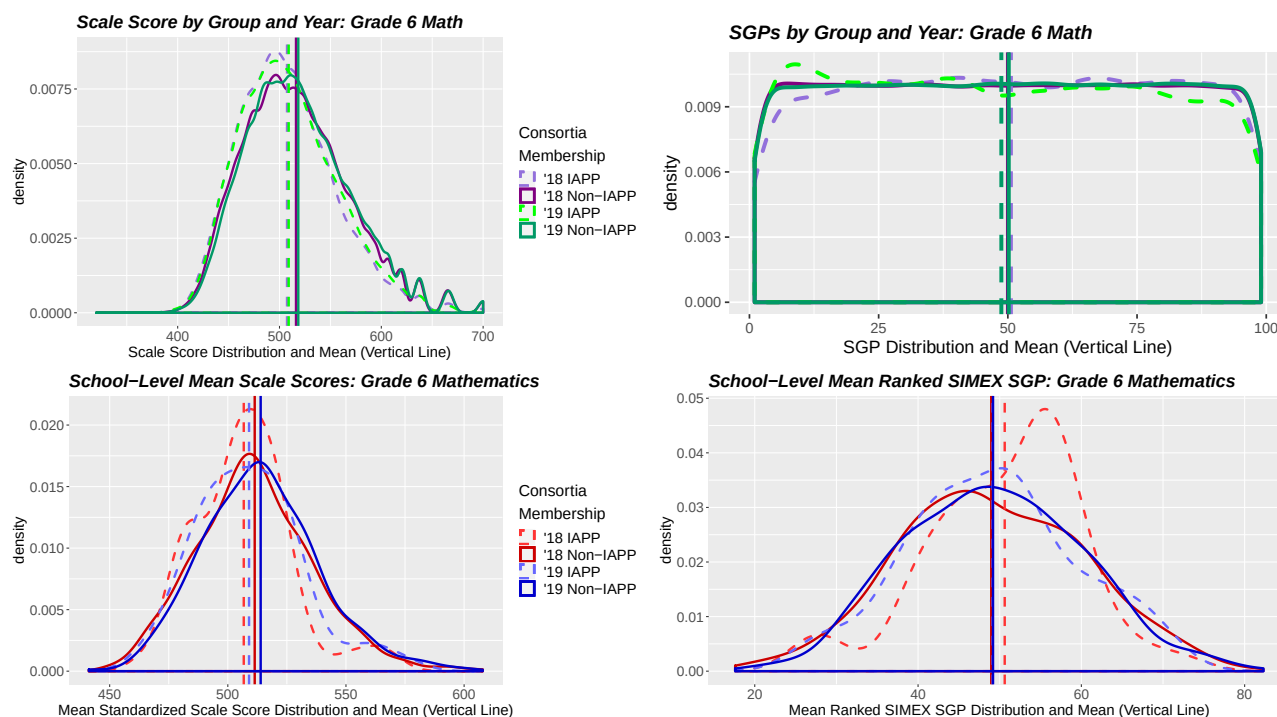


Table 29: Mean and Median Values for 6th Grade Math - Complete Georgia Analysis.

Group	Year	Scale Scores		SGPs		N
		Mean	Median	Mean	Median	
IAPP Students	2018	507.4	504.0	50.9	51.0	13,105
	2019	509.0	504.0	49.0	49.0	13,412
Non-IAPP Students	2018	516.2	511.0	50.0	50.0	111,660
	2019	518.5	513.0	50.2	50.0	114,766
IAPP School Means	2018	506.7	502.5	50.6	53.2	52
	2019	509.0	504.0	49.1	48.0	52
Non-IAPP School Means	2018	511.4	507.0	49.0	47.5	515
	2019	513.9	510.0	49.2	48.5	513

6.6 6th Grade Math IAPP Consortium Exclusion Comparison

Table 30: Differences in Student Level 6th Grade Math SGPs (Non-IAPP Results minus Complete).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-21	-1	0	0.06	1	17
2019	-48	-1	0	-0.13	1	47

Figure 20: 2018 and 2019 6th Grade Math Alternate Analysis Comparison - School Level Mean SGPs (MSGP)

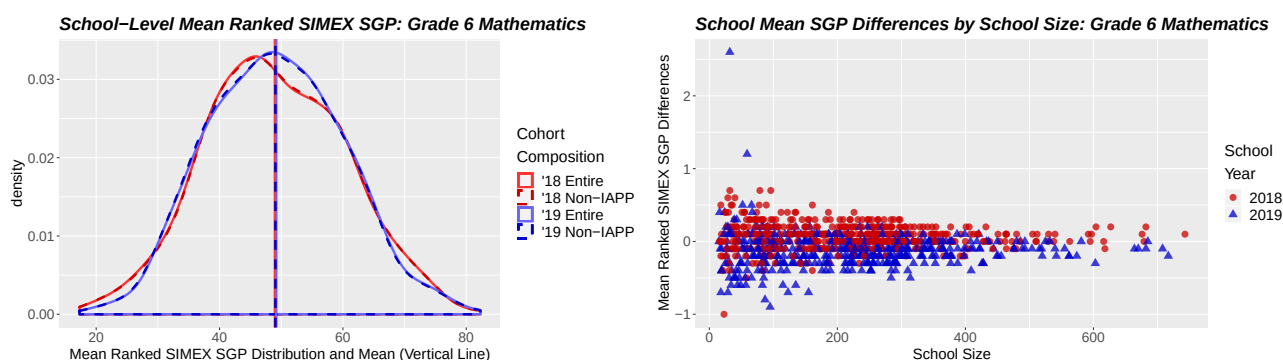


Table 31: Differences in School Level 6th Grade Math Mean SGPs (Non-IAPP District MSGP minus Complete MSGP).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-1.0	0.0	0.1	0.07	0.2	0.7
2019	-0.9	-0.2	-0.1	-0.13	0.0	2.6

The IAPP students and schools show overall lower achievement in Grade 6 Math and their growth is similar (with the IAPP groups in outperforming slightly in 2018). Although omitting IAPP students from the SGP norm group impacts some individual students' growth estimates substantially, there is minimal change in the school-level MSGPs. Differences in these aggregates range from approximately -1 to 2.5, which is much lower than annual fluctuations in MSGPs for most schools. The impact of removing IAPP students also decreases substantially with increasing school size.

6.7 7th Grade Mathematics by Consortium Membership

Figure 21: 2018 and 2019 7th Grade Mathematics Consortium Member Comparisons

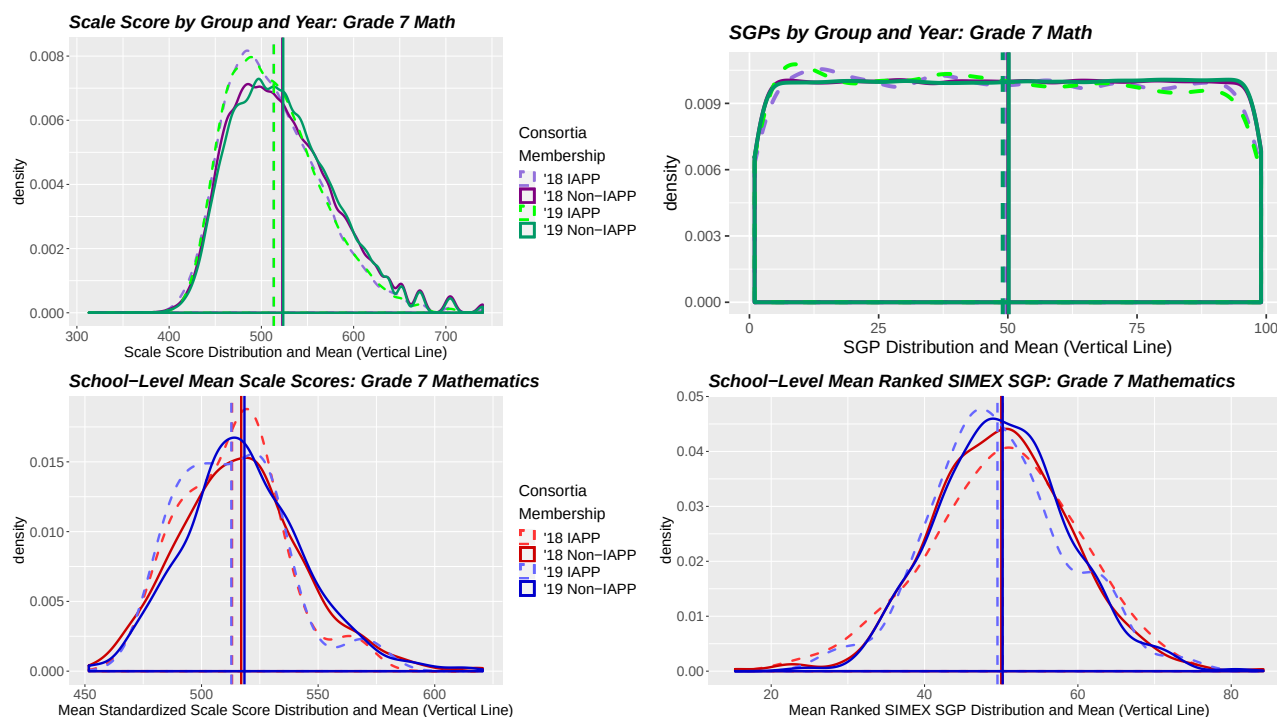


Table 32: Mean and Median Values for 7th Grade Math - Complete Georgia Analysis.

Group	Year	Scale Scores		SGPs		N
		Mean	Median	Mean	Median	
IAPP Students	2018	513.4	507.0	49.8	50.0	12,659
	2019	513.5	507.0	49.4	49.0	13,046
Non-IAPP Students	2018	522.7	515.0	50.1	50.0	109,127
	2019	523.7	517.0	50.2	50.0	111,548
IAPP School Means	2018	513.1	509.0	50.2	51.2	52
	2019	512.8	507.0	49.5	47.5	52
Non-IAPP School Means	2018	517.0	511.0	50.0	50.0	494
	2019	518.4	511.5	50.2	50.0	501

6.8 7th Grade Math IAPP Consortium Exclusion Comparison

Table 33: Differences in Student Level 7th Grade Math SGPs (Non-IAPP Results minus Complete).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-33	-1	0	-0.07	1	18
2019	-51	-1	0	-0.10	1	46

Figure 22: 2018 and 2019 7th Grade Math Alternate Analysis Comparison - School Level Mean SGPs (MSGP)

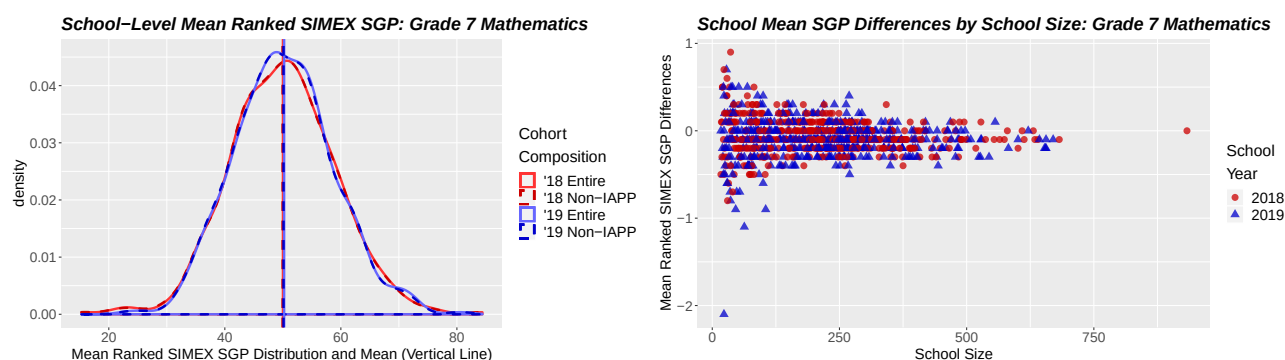


Table 34: Differences in School Level 7th Grade Math Mean SGPs (Non-IAPP District MSGP minus Complete MSGP).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-0.8	-0.2	-0.1	-0.06	0	0.9
2019	-2.1	-0.2	-0.1	-0.10	0	0.7

The IAPP students and schools show overall lower achievement in Grade 7 Math and their growth is similar. Although omitting IAPP students from the SGP norm group impacts some individual students' growth estimates substantially, there is minimal change in the school-level MSGPs. Differences in these aggregates range from approximately -2 to 1, which is much lower than annual fluctuations in MSGPs for most schools. The impact of removing IAPP students also decreases substantially with increasing school size.

6.9 8th Grade Mathematics by Consortium Membership

Figure 23: 2018 and 2019 8th Grade Mathematics Consortium Member Comparisons

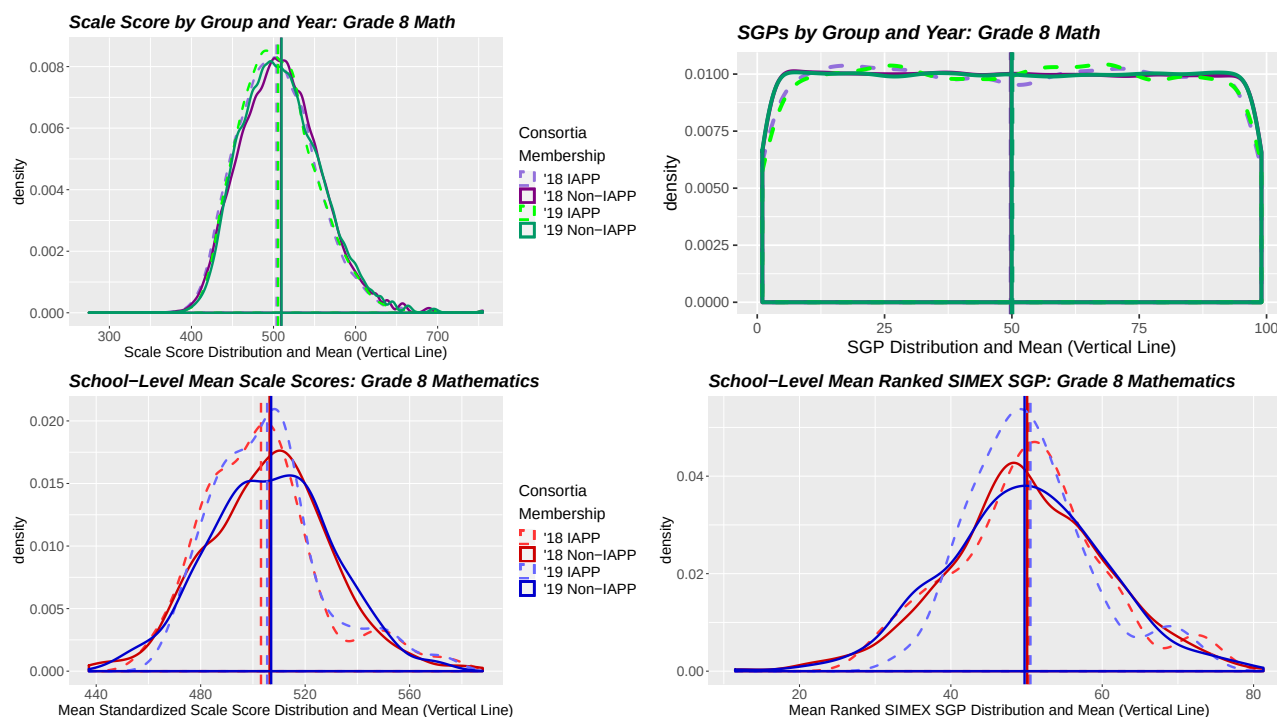


Table 35: Mean and Median Values for 8th Grade Math - Complete Georgia Analysis.

Group	Year	Scale Scores		SGPs		N
		Mean	Median	Mean	Median	
IAPP Students	2018	503.8	501	50.0	50	10,855
	2019	505.6	502	50.3	51	10,956
Non-IAPP Students	2018	509.3	506	49.9	50	86,950
	2019	509.7	507	50.0	50	85,367
IAPP School Means	2018	503.1	505	50.3	51	51
	2019	505.4	502	50.5	50	51
Non-IAPP School Means	2018	506.4	506	50.1	49	487
	2019	507.0	506	49.7	50	491

6.10 8th Grade Math IAPP Consortium Exclusion Comparison

Table 36: Differences in Student Level 8th Grade Math SGPs (Non-IAPP Results minus Complete).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-11	-1	0	-0.01	1	12
2019	-46	-1	0	-0.02	1	49

Figure 24: 2018 and 2019 8th Grade Math Alternate Analysis Comparison - School Level Mean SGPs (MSGP)

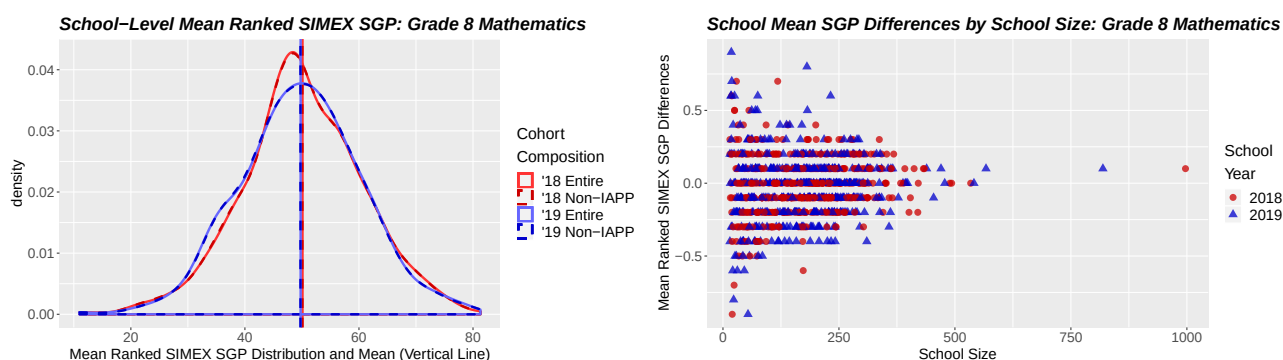


Table 37: Differences in School Level 8th Grade Math Mean SGPs (Non-IAPP District MSGP minus Complete MSGP).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-0.9	-0.1	0	-0.02	0.1	0.7
2019	-0.9	-0.2	0	-0.03	0.1	0.9

The IAPP students and schools show slightly lower achievement in Grade 8 Math and their growth is similar (IAPP group slightly higher). Although omitting IAPP students from the SGP norm group impacts some individual students' growth estimates substantially, there is minimal change in the school-level MSGPs. Differences in these aggregates range from approximately -1 to 1, which is much lower than annual fluctuations in MSGPs for most schools. The impact of removing IAPP students also decreases substantially with increasing school size.

7 End-of-Course Test Results

In reviewing the end-of-course assessment results, it must be noted that they are collapsed across all course progressions, making them look more uneven than the grade level assessments. However, because schools are consistent in their annual use of the same course progressions, the aggregation of these results should yield consistent patterns for the IAPP and Non-IAPP.

7.1 Grade 9 Lit by Consortium Membership

Figure 25: 2018 and 2019 Grade 9 Lit Consortium Member Comparisons

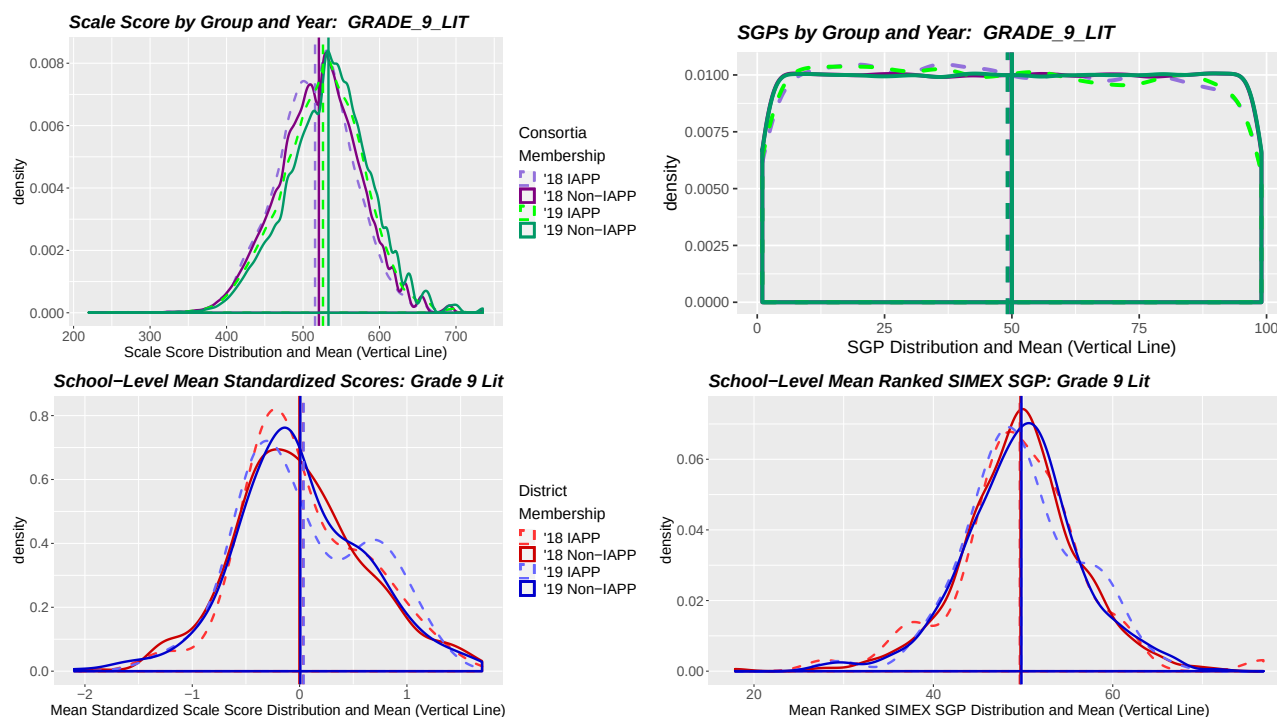


Table 38: Mean and Median Values for Grade 9 Lit - Complete Georgia Analysis.

Group	Year	Scale Scores		SGPs		N
		Mean	Median	Mean	Median	
IAPP Students	2018	515.7	516.0	49.6	49.0	12,989
	2019	526.1	528.0	49.5	49.0	12,626
Non-IAPP Students	2018	520.7	525.0	50.0	50.0	115,359
	2019	533.3	534.0	50.1	50.0	114,502
IAPP School Means	2018	522.5	516.0	49.6	49.0	59
	2019	534.7	529.5	49.8	49.5	60
Non-IAPP School Means	2018	520.0	520.0	49.7	49.5	468
	2019	532.9	530.0	49.8	49.5	487

7.2 Grade 9 Lit IAPP Consortium Exclusion Comparison

Table 39: Differences in Student Level Grade 9 Lit SGPs (Non-IAPP Results minus Complete).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-62	-1	0	-0.07	1	53
2019	-61	-1	0	-0.07	1	50

Figure 26: 2018 and 2019 Grade 9 Lit Alternate Analysis Comparison - School Level Mean SGPs (MSGP)

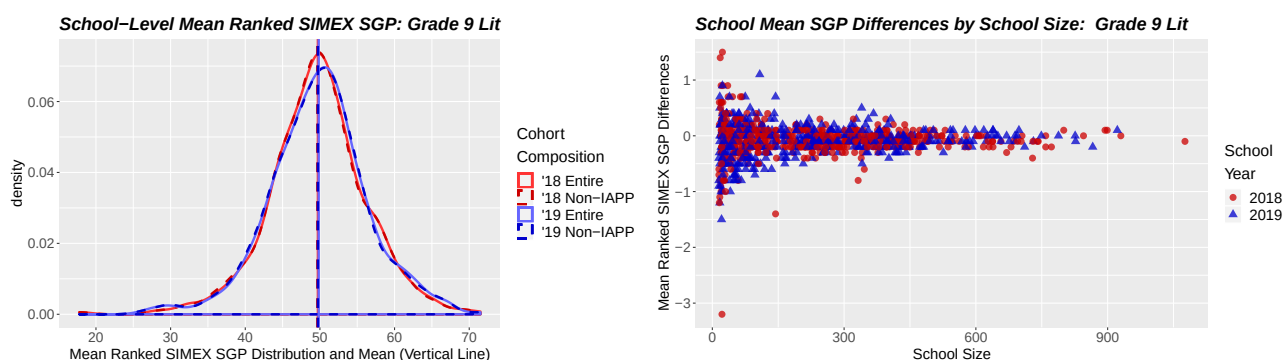


Table 40: Differences in School Level Grade 9 Lit Mean SGPs (Non-IAPP District MSGP minus Complete MSGP).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-3.2	-0.2	-0.1	-0.07	0.1	1.5
2019	-1.5	-0.2	-0.1	-0.11	0.0	1.1

The IAPP students show lower achievement in Grade 9 Literature, while schools averages are slightly higher. Their growth is similar for both groups, particularly at the school level. Although omitting IAPP students from the SGP norm group impacts some individual students' growth estimates substantially, there is minimal change in the school-level MSGPs. Differences in these aggregates range from approximately -3 to 1.5, which is much lower than annual fluctuations in MSGPs for most schools. The impact of removing IAPP students also decreases substantially with increasing school size.

7.3 American Lit by Consortium Membership

Figure 27: 2018 and 2019 American Lit Consortium Member Comparisons

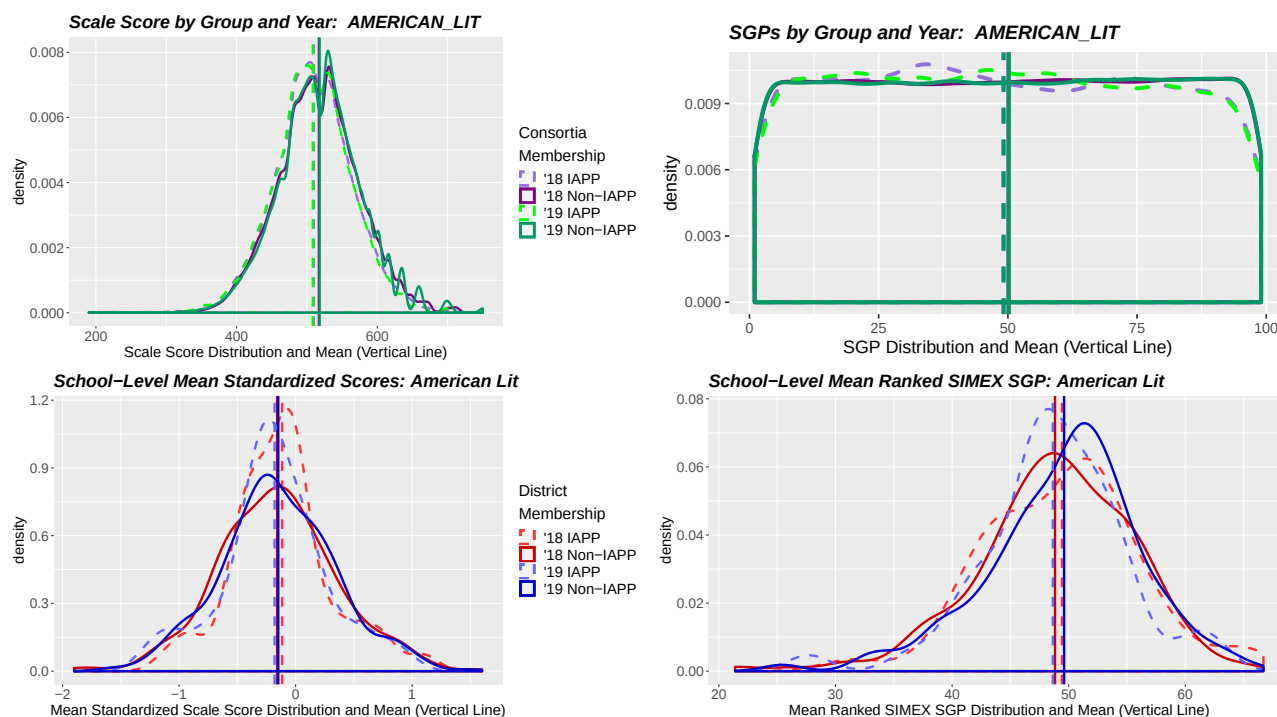


Table 41: Mean and Median Values for American Lit - Complete Georgia Analysis.

Group	Year	Scale Scores		SGPs		N
		Mean	Median	Mean	Median	
IAPP Students	2018	509.6	510	49.6	49.0	11,831
	2019	508.3	507	49.7	50.0	11,551
Non-IAPP Students	2018	517.0	515	50.2	50.0	105,224
	2019	518.0	516	50.3	50.0	104,810
IAPP School Means	2018	509.5	510	49.4	50.5	45
	2019	506.7	505	48.6	49.0	46
Non-IAPP School Means	2018	507.2	507	48.8	49.0	389
	2019	508.4	505	49.6	50.0	391

7.4 American Lit IAPP Consortium Exclusion Comparison

Table 42: Differences in Student Level American Lit SGPs (Non-IAPP Results minus Complete).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-27	-1	0	-0.06	1	49
2019	-42	-1	0	-0.07	1	40

Figure 28: 2018 and 2019 American Lit Alternate Analysis Comparison - School Level Mean SGPs (MSGP)

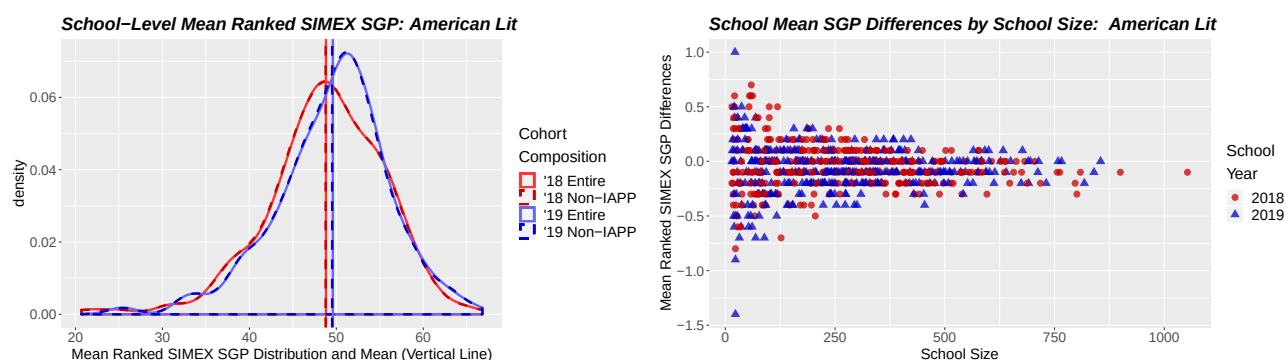


Table 43: Differences in School Level American Lit Mean SGPs (Non-IAPP District MSGP minus Complete MSGP).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-0.8	-0.1	-0.1	-0.04	0.1	0.7
2019	-1.4	-0.2	-0.1	-0.08	0.0	1.0

The IAPP students show lower achievement in American Literature, while schools averages are similar. Their growth is similar for both groups, particularly at the school level. Although omitting IAPP students from the SGP norm group impacts some individual students' growth estimates substantially, there is minimal change in the school-level MSGPs. Differences in these aggregates range from approximately -1.5 to 1, which is much lower than annual fluctuations in MSGPs for most schools. The impact of removing IAPP students also decreases substantially with increasing school size.

7.5 Algebra I by Consortium Membership

Figure 29: 2018 and 2019 Algebra I Consortium Member Comparisons

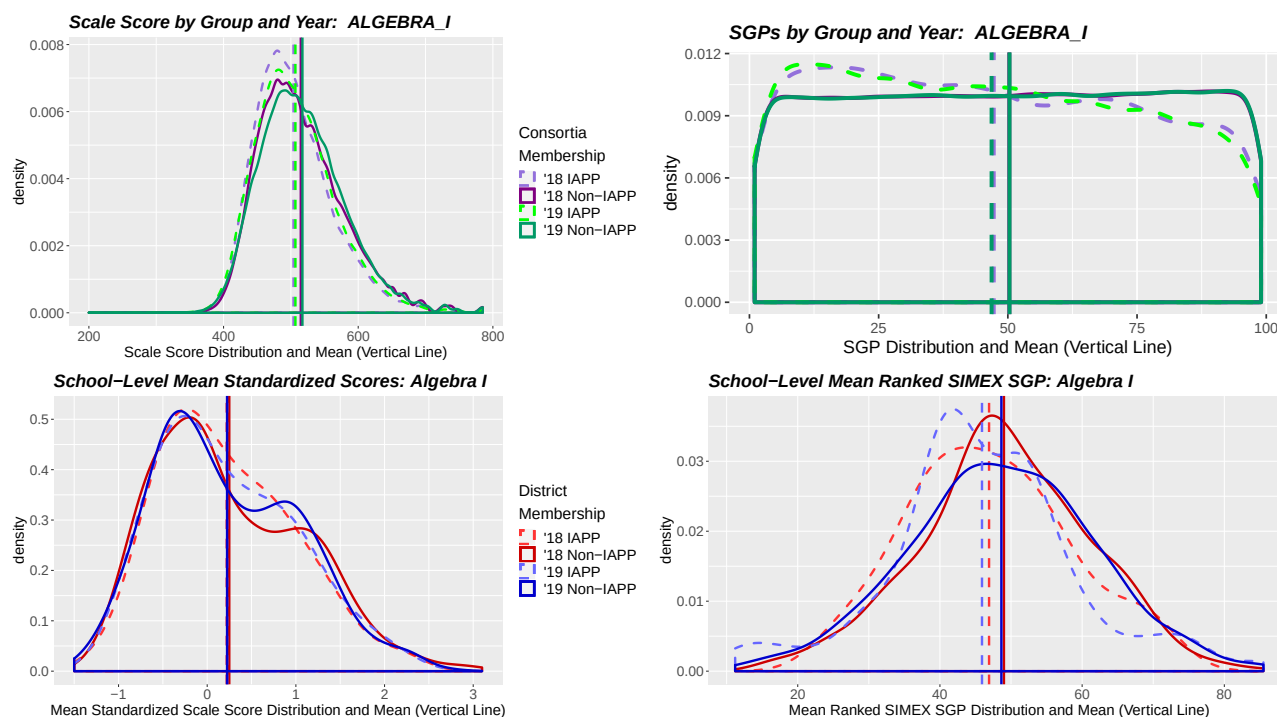


Table 44: Mean and Median Values for Algebra I - Complete Georgia Analysis.

Group	Year	Scale Scores		SGPs		N
		Mean	Median	Mean	Median	
IAPP Students	2018	503.9	497.0	47.5	46.0	12,544
	2019	506.7	499.0	47.0	46.0	12,282
Non-IAPP Students	2018	514.7	506.0	50.3	51.0	101,057
	2019	517.2	510.0	50.4	51.0	101,035
IAPP School Means	2018	527.4	511.0	46.9	44.5	80
	2019	530.1	518.0	45.9	43.0	81
Non-IAPP School Means	2018	529.1	513.5	49.0	48.0	566
	2019	530.1	518.5	48.6	48.0	594

7.6 Algebra I IAPP Consortium Exclusion Comparison

Table 45: Differences in Student Level Algebra I SGPs (Non-IAPP Results minus Complete).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-71	-1	0	-0.32	1	45
2019	-47	-1	0	-0.40	0	43

Figure 30: 2018 and 2019 Algebra I Alternate Analysis Comparison - School Level Mean SGPs (MSGP)

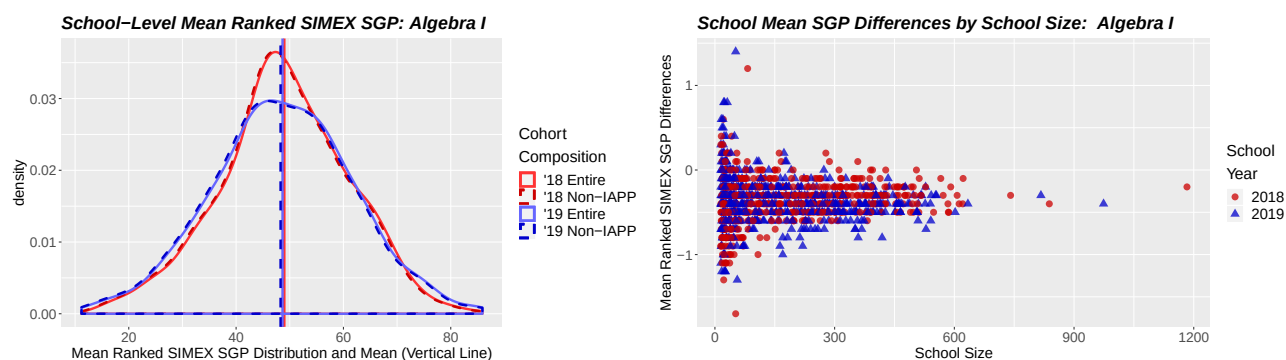


Table 46: Differences in School Level Algebra I Mean SGPs (Non-IAPP District MSGP minus Complete MSGP).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-1.7	-0.5	-0.3	-0.35	-0.2	1.2
2019	-1.3	-0.5	-0.4	-0.37	-0.2	1.4

The IAPP students show lower achievement in Algebra I, while schools averages are only slightly lower. IAPP students and schools also underperform in growth. Although omitting IAPP students from the SGP norm group impacts some individual students' growth estimates substantially, there is minimal change in the school-level MSGPs. Differences in these aggregates range from approximately -1.5 to 1.5, which is much lower than annual fluctuations in MSGPs for most schools. The impact of removing IAPP students also decreases substantially with increasing school size.

7.7 Geometry by Consortium Membership

Figure 31: 2018 and 2019 Geometry Consortium Member Comparisons



Table 47: Mean and Median Values for Geometry - Complete Georgia Analysis.

Group	Year	Scale Scores		SGPs		N
		Mean	Median	Mean	Median	
IAPP Students	2018	512.9	504.0	50.4	51.0	10,882
	2019	512.9	503.0	50.6	51.0	11,560
Non-IAPP Students	2018	522.5	513.0	49.3	49.0	90,179
	2019	524.6	517.0	49.2	49.0	93,499
IAPP School Means	2018	513.3	503.0	49.7	50.5	41
	2019	511.9	503.5	50.2	51.0	42
Non-IAPP School Means	2018	514.9	507.0	48.8	49.0	307
	2019	516.2	505.0	48.4	47.0	332

7.8 Geometry IAPP Consortium Exclusion Comparison

Table 48: Differences in Student Level Geometry SGPs (Non-IAPP Results minus Complete).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-68	-1	0	0.12	1	44
2019	-49	-1	0	0.19	1	54

Figure 32: 2018 and 2019 Geometry Alternate Analysis Comparison - School Level Mean SGPs (MSGP)

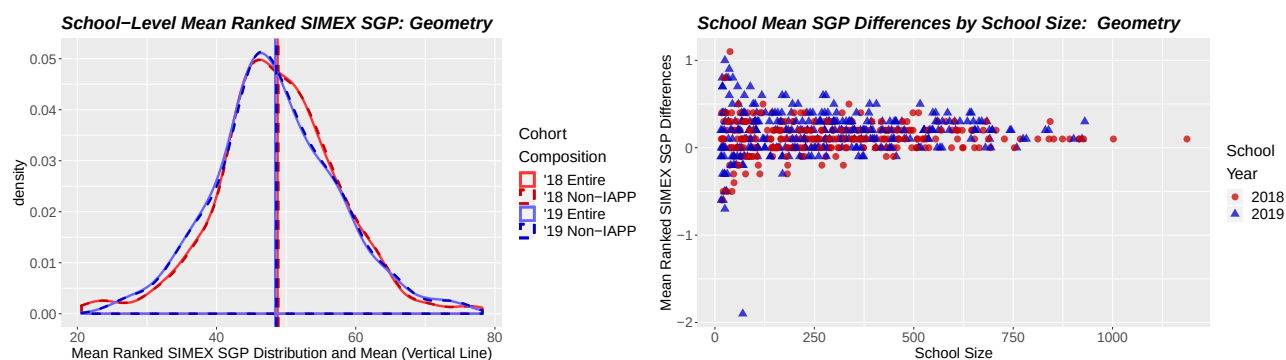


Table 49: Differences in School Level Geometry Mean SGPs (Non-IAPP District MSGP minus Complete MSGP).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-0.6	0	0.1	0.12	0.2	1.1
2019	-1.9	0	0.2	0.16	0.3	1.0

The IAPP students show lower achievement in Geometry, while schools averages are only slightly lower. However, IAPP students and schools outperform slightly in growth. Although omitting IAPP students from the SGP norm group impacts some individual students' growth estimates substantially, there is minimal change in the school-level MSGPs. Differences in these aggregates range from approximately -2 to 1, which is much lower than annual fluctuations in MSGPs for most schools. The impact of removing IAPP students also decreases substantially with increasing school size.

7.9 Coordinate Algebra by Consortium Membership

Figure 33: 2018 and 2019 Coordinate Algebra Consortium Member Comparisons

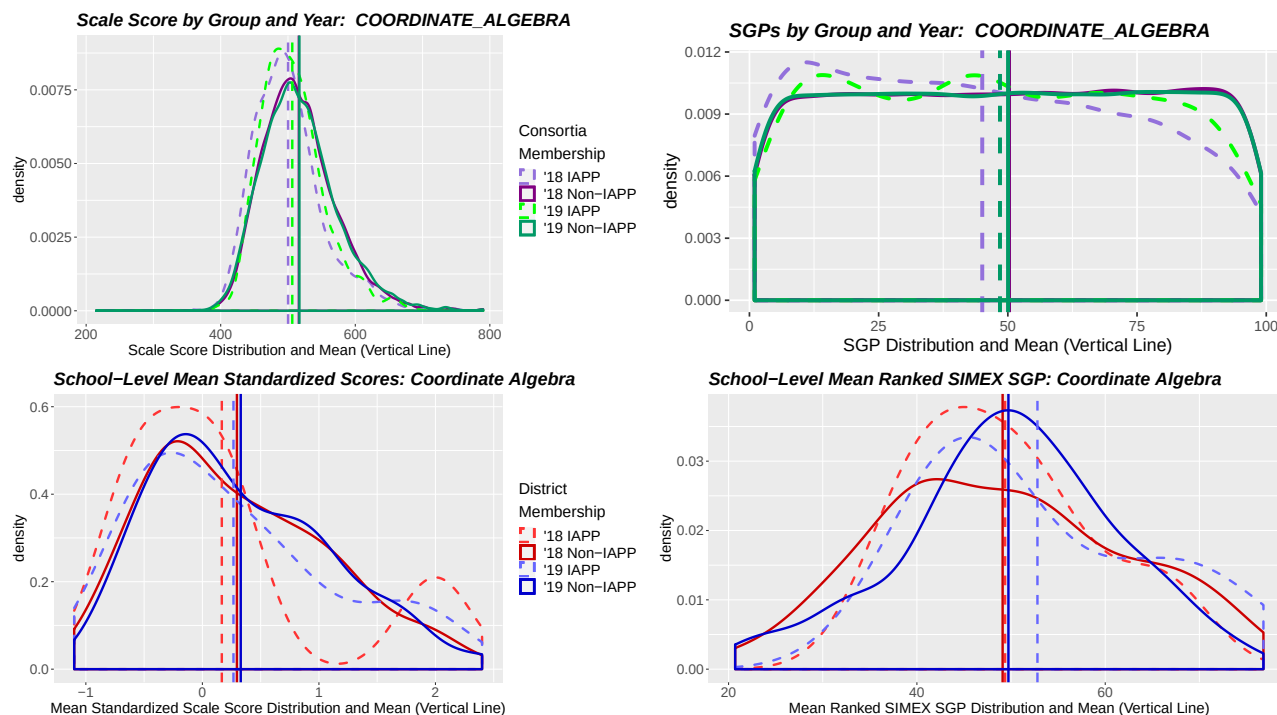


Table 50: Mean and Median Values for Coordinate Algebra - Complete Georgia Analysis.

Group	Year	Scale Scores		SGPs		N
		Mean	Median	Mean	Median	
IAPP Students	2018	500.1	494.0	45.5	44.0	823
	2019	506.4	500.0	48.5	47.0	722
Non-IAPP Students	2018	516.1	510.0	50.4	51.0	15,394
	2019	517.0	512.0	50.2	50.0	14,832
IAPP School Means	2018	523.4	508.8	49.4	50.0	6
	2019	531.6	514.8	52.8	47.2	6
Non-IAPP School Means	2018	531.2	521.8	49.1	49.0	108
	2019	534.6	521.0	49.7	50.0	103

7.10 Coordinate Algebra IAPP Consortium Exclusion Comparison

Table 51: Differences in Student Level Coordinate Algebra SGPs (Non-IAPP Results minus Complete).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-51	-2	0	-0.30	1	41
2019	-35	-1	0	-0.09	1	29

Figure 34: 2018 and 2019 Coordinate Algebra Alternate Analysis Comparison - School Level Mean SGPs (MSGP)

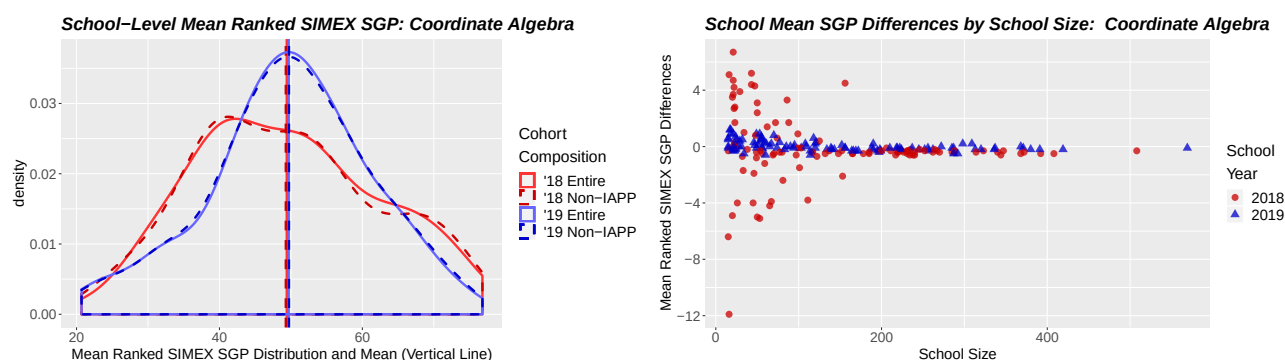


Table 52: Differences in School Level Coordinate Algebra Mean SGPs (Non-IAPP District MSGP minus Complete MSGP).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-11.9	-0.5	-0.4	-0.13	0.05	6.7
2019	-0.6	-0.2	-0.1	0.04	0.20	1.2

When comparing the Coordinate Algebra results it is important to note that there are only six IAPP schools in this cohort.

The IAPP students and schools show lower achievement in Coordinate Algebra. IAPP students underperform in growth, while schools are generally similar in growth (note the difference between mean and median MSGPs for 2019 IAPP schools). As with other subjects, omitting IAPP students from the SGP norm group impacts some individual students' growth estimates substantially. There is also greater change in the school-level MSGPs. Differences in these aggregates range from approximately -12 to 7 in 2018, and -0.5 to 1 in 2019, which is still lower than annual fluctuations in MSGPs for these schools (which ranges from -25 to 22 for

non-IAPP schools from 2018 to 2019). The impact of removing IAPP students also decreases substantially with increasing school size.

7.11 Analytic Geometry by Consortium Membership

Figure 35: 2018 and 2019 Analytic Geometry Consortium Member Comparisons

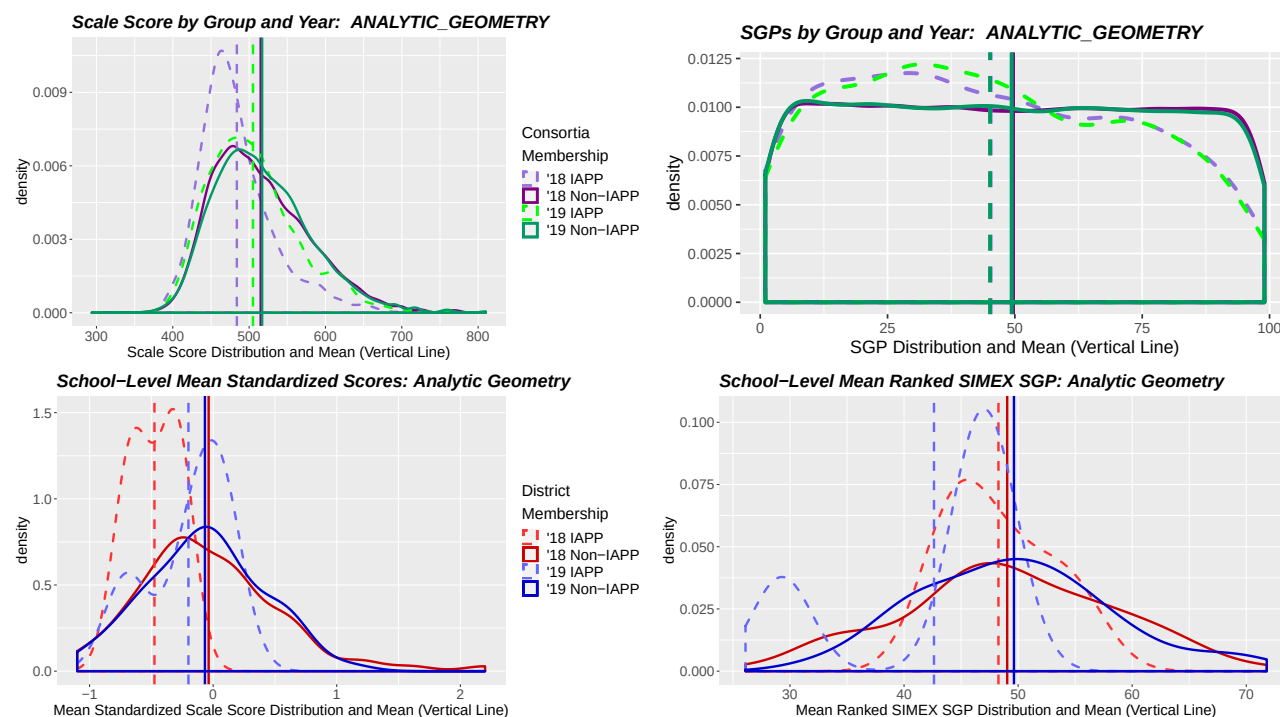


Table 53: Mean and Median Values for Analytic Geometry - Complete Georgia Analysis.

Group	Year	Scale Scores		SGPs		N
		Mean	Median	Mean	Median	
IAPP Students	2018	483.9	474.0	47.2	46.5	1,010
	2019	505.1	495.0	45.9	44.0	658
Non-IAPP Students	2018	515.1	504.0	50.0	50.0	19,553
	2019	517.1	509.0	49.7	50.0	14,946
IAPP School Means	2018	484.9	475.0	48.3	45.0	4
	2019	504.2	505.8	42.6	43.8	4
Non-IAPP School Means	2018	511.1	498.0	49.0	47.0	85
	2019	512.1	504.0	49.6	48.5	63

7.12 Analytic Geometry IAPP Consortium Exclusion Comparison

Table 54: Differences in Student Level Analytic Geometry SGPs (Non-IAPP Results minus Complete).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-55	-1	0	-0.16	1	17
2019	-33	-1	0	-0.12	1	33

Figure 36: 2018 and 2019 Analytic Geometry Alternate Analysis Comparison - School Level Mean SGPs (MSGP)

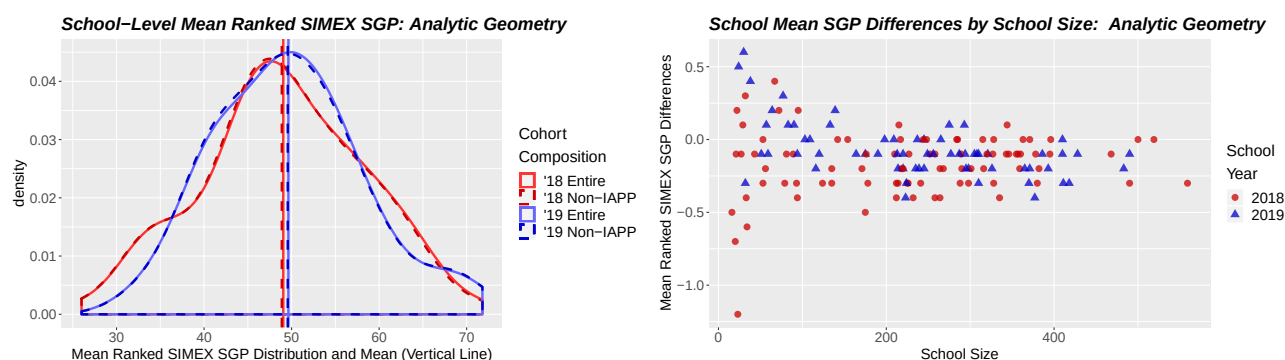


Table 55: Differences in School Level Analytic Geometry Mean SGPs (Non-IAPP District MSGP minus Complete MSGP).

School Year	Min	1st Qu	Median	Mean	3rd Qu	Max
2018	-1.2	-0.3	-0.1	-0.17	0	0.4
2019	-0.4	-0.2	-0.1	-0.07	0	0.6

When comparing the Analytic Geometry results it is important to note that there are only four IAPP schools in this cohort.

The IAPP students and schools show lower achievement in Analytic Geometry. IAPP students and schools both underperform in growth, while schools are generally similar in growth, particularly in 2019. Although omitting IAPP students from the SGP norm group impacts some individual students' growth estimates substantially, there is minimal change in the school-level MSGPs. Differences in these aggregates range from approximately -1 to 0.5, which is much lower than annual fluctuations in MSGPs for most schools. The impact of removing IAPP students also decreases substantially with increasing school size.

8 Discussion

This exploratory analysis looked at both the comparability of IAPP and non-IAPP students and schools, as well as the differences that Georgia might expect to see if those schools are removed from the cohorts from which SGPs are calculated. At the demographic level, IAPP consortia have more students that are economically disadvantaged. Overall they tend to be lower performing academically at the student level, and this typically results in lower school-level average academic achievement. Although growth is relatively similar in most cases, the IAPP students and schools also tend to have somewhat lower growth on average.

Due to these factors, removing these students from the growth calculation cohorts has an overall negative impact. However, this impact is small when compared to the annual fluctuation in MSGPs at the school level. For example, when looking at the change in schools (again with a minimum of 15 students) in their MSGPs from 2018 to 2019, the range in differences is as large as -45 to 50. For most subjects and grades in the analyses excluding the IAPP consortia, the differences were within the range of -3 to 3 (with the exception of 2018 Coordinate Algebra).

Although this exploratory analysis does suggest some impact to be expected from removing the IAPP consortium schools from growth analyses, particularly in individual student estimates, the changes do not appear to be significant enough to prevent Georgia from moving forward with their IADA pilot program. However, more in depth analyses may be required going forward.